

Jianfeng Cai

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Purdue University
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EDUCATION

- **Postdoctoral Associate**, Bioorganic Chemistry, Yale University, **2007-2009**
Advisor: **Professor Andrew D. Hamilton**
- **PhD**, Bioorganic Chemistry, Washington University in St. Louis, **2002-2006**
Advisor: **Professor John-Stephen Taylor**
Thesis Title: *Design and Synthesis of Nucleic Acid Templated and Targeted Drugs and Probes*
- **MS**, Nanjing University, China, **2000**
- **BS**, Nanjing University, China, **1997**

POSITIONS AND EMPLOYMENT

- 2007-2009 Postdoctoral Associate, Yale University, New Haven, CT
- 2009-2015 Assistant Professor, University of South Florida, Tampa, FL
- 2015-2018 Associate Professor, University of South Florida, Tampa, FL
- 2018-2026 Professor, University of South Florida, Tampa, FL
- 2009-2026 Member, Drug Discovery Program, Moffitt Cancer Center, Tampa, FL
- 2019-2026 Director, Center for Molecular Diversity in Drug Design, Discovery and Development (CMD5)
- 2020-2026 USF Preeminent Professor, University of South Florida, Tampa, FL
- 2024-2026 Distinguished University Professor, University of South Florida, Tampa, FL
- 2025-2026 Julie Harmon Endowed Chair in Chemistry
- 2026-Present Professor, Purdue University, West Lafayette, IN

RESEARCH INTEREST

Research Area: Bioorganic, Chemical Biology, Medicinal Chemistry, Biophysics, and Biomaterials

Research Focus: Design, synthesis and investigation of AApeptide-based bioactive peptidomimetics; development of novel biomaterials

AWARDS AND RECOGNITIONS

- 2025 Member, Academy of Science, Engineering and Medicine of Florida (ASEMFL)
- 2023-2025 World's Top 2% Scientists, Stanford/Elsevier
- 2025 Julie Harmon Endowed Chair in Chemistry
- 2024 Fellow of the American Association for the Advancement of Science (AAAS)
- 2024 Fellow of National Academy of Inventors (NAI)
- 2024 USF Faculty Outstanding Research Achievement Award
- 2024 Distinguished University Professor

2024	OKeanos-CAPA Senior Investigator Award at the Chemical and Biology Interface
2023	The Huber and Helen Croft Lectureship, University of Missouri
2023	Fellow of American Institute for Medical and Biological Engineering (AIMBE)
2021	Outstanding Faculty Award, USF
2020	Outstanding Graduate Faculty Mentor Award, USF
2020	USF Faculty Outstanding Research Achievement Award
2020	USF Preeminent Professor
2020	Fellow of Royal Society of Chemistry (FRSC)
2018	USF Faculty Outstanding Research Achievement Award
2015	USF Faculty Outstanding Research Achievement Award
2015	Biomatik Distinguished Junior Faculty Award, the Chinese-American Chemistry & Chemical Biology Professors Association (CAPA)
2014	NSF Career Award
2014	ChemComm Emerging Investigator
2012	New Investigator award, Florida Bankhead Coley Cancer Research Program
2011	Ralph E. Powe Junior Faculty Enhancement Award, Oak Ridge Associated Universities

PROFESSIONAL MEMBERSHIPS

Member, American Chemical Society (Organic Chemistry and Medicinal Chemistry Division)

Member, American Peptide Society

Member, National Academy of Inventors, USF Chapter

PROFESSIONAL SERVICES

Editorial and Consulting Services:

2015-	Editorial Board member, <i>ChemistrySelect</i>
2017-	Editorial Advisory Board member, <i>ChemistryOpen</i>
2018-	Associate Editor, Chemical Biology Section, <i>Molecules</i>
2020-2023	Associate Editor-in-Chief, <i>Acta Pharmaceutica Sinica B</i>
2023-	Consultant, Fulgent Pharma LLC
2024-	Executive Board member and Vice President, USF Chapter of National Academy of Inventors
2024-	Associate Editor, <i>J. Enzyme Inhib. Med. Chem.</i>
2025-	Co-founder, MolAlcule

Grant Review Services:

2015.4	Panelist, CHEM-CLP, National Science Foundation
2015.6	Ad hoc member, BMBI, National Institute of Health
2017.2	Ad hoc member, SBCB, National Institute of Health
2017.7	Ad hoc member, Special Emphasis Panel, ZRG1 IDM-S (02) M, National Institute of Health
2017.11	Ad hoc member, Special Emphasis Panel, ZAI1 LG-M (J1), 1 National Institute of Health
2017.11	Ad hoc member, BMBI, ZRG1 BST-M (90) S, National Institute of Health
2018.3	Ad hoc member, Special Emphasis Panel, ZRG1 IDM-Y 82, National Institute of Health
2018. 9	Panelist, CHEM-CLP, National Science Foundation
2018. 10	Ad hoc member, Special Emphasis Panel, ZAG1 ZIJ-7 (J1), National Institute of Health
2018.11	Ad hoc member, Special Emphasis Panel, ZRG1 IDM-Y 82, National Institute of Health
2019.2	Ad hoc member, EBIT, National Institute of Health

2020.1	Ad hoc member, BMBI, National Institute of Health
2020.2	Ad hoc member, EBIT, National Institute of Health
2020.3	Panelist, BMAT, National Science Foundation
2020.6-2024	Standing member, BMBI, National Institute of Health
2021.4	Panelist, BMAT, National Science Foundation
2022.2	NIH Avid study section
2023.1	NSF ad hoc reviewer
2023.4	NSF ad hoc reviewer
2023.10	NSF ad hoc reviewer
2024. 8	NIH ad hoc reviewer

PUBLICATIONS

Selected representative publications:

19. Heng Liu, Xue Zhao, Qiao Qiao, Lukasz Wojtas and **Jianfeng Cai**.^{*} Sulfonyl- γ -AAs as Turn Templates Inducing β -Sheet Conformation in Macrocyclic Peptides. *J. Am. Chem. Soc.* 2026, ASAP.
18. Heng Liu,⁺ Firoz Akhter,⁺ Asma Akhter,⁺ Xue Zhao, Xudong Wang, Haiqiang Yang, Jianyu Chen, Xinyu Xia, Laurent Calcul, Yi He, Chuanhai Cao, Libin Ye, Donghui Zhu,^{*} and **Jianfeng Cai**.^{*} α /Sulfonyl- γ -AApeptide Foldamers Mitigate Alzheimer's Disease Pathology by Stabilizing Transient Helical Domains in A β . *Nat. Commun.* 2026, ASAP.
17. Peng Sang,^{*,+} Jiacheng Wei,⁺ Mingshuang Wu, Qingqing Li, Jian Shang, Baoguo Li, Jian Zhang,^{*} **Jianfeng Cai**,^{*} and Yan Shi.^{*} Disruption of β -catenin/B Cell Lymphoma 9 Protein-Protein Interaction Using Heterogeneous Peptidomimetic Foldamers. *J. Am. Chem. Soc.* 2025, 147, 41819-41829.
16. Bo Huang, Sihao Li, Cong Pan, Fangzhou Li, Lukasz Wojtas, Qiao Qiao, Timothy H. Tran, Laurent Calcul, Wenqi Liu, Chenfeng Ke,^{*} **Jianfeng Cai**.^{*} Proline-Based Tripodal Cages with Guest-Adaptive Features for Capturing Hydrophilic and Amphiphilic Fluoride Substances. *Nat. Commun.* 2025, 16, 3226.
15. Heng Liu, Yunpeng Cui, Xue Zhao, Lulu Wei, Xudong Wang, Ning Shen, Timothy Odom, Xuming Li, William Lawless, Kanchana Karunaratne, Martin Muschol, Wayne Guida, Chuanhai Cao, Libin Ye, and **Jianfeng Cai**.^{*} Helical sulfonyl- γ -AApeptides modulating A β oligomerization and cytotoxicity by recognizing A β helix. *Proc. Natl. Acad. Sci. U. S. A.* 2024, 121, 6, e2311733121.
14. Wei Jiang,⁺ Sami Abdulkadir,⁺ Xue Zhao, Peng Sang, Laurent Calcul, Feng Cheng, ^{*} Yong Hu, ^{*} and **Jianfeng Cai**.^{*} Inhibition of Hypoxia-Inducible Transcription Factor (HIF-1 α) with Sulfonyl- γ -AApeptide Helices. *J. Am. Chem. Soc.*, 2023, 145, 36, 20009-20020.
13. Songyi Xue,⁺ Wei Xu,^{+,*} Lei Wang, Xinling Wang, Qianyu Duan, Laurent Calcul, Shaohui Wang, Wenqi Liu, Xingmin Sun, Lu Lu,^{*} Shibo Jiang,^{*} and **Jianfeng Cai**.^{*} An HR2-mimicking sulfonyl- γ -AApeptide is a potent pan-coronavirus fusion inhibitor with strong blood-brain barrier permeability, long half-life and promising oral bioavailability. *ACS Central. Sci.* 2023, 9 1046-1058.
12. Sami Abdulkadir,⁺ Chunpu Li,⁺ Wei Jiang,⁺ Xue Zhao, Peng Sang, Lulu Wei, Yong Hu,^{*} Qi Li,^{*} and **Jianfeng Cai**.^{*} Modulating Angiogenesis by Proteomimetics of Vascular Endothelial Growth Factor. *J. Am. Chem. Soc.*, 2022, 144, 1, 270–281.
11. Songyi Xue,⁺ Xinling Wang,⁺ Lei Wang, Wei Xu, Shuai Xia, Lujia Sun, Shaohui Wang, Ning Shen, Ziqi Yang, Bo Huang, Sihao Li, Chuanhai Cao, Laurent Calcul, Xingmin Sun, Lu Lu,^{*} **Jianfeng Cai**,^{*} and Shibo Jiang.^{*} A novel cyclic $\hat{I}^3$ -AApeptide-based long-acting pan-coronavirus fusion inhibitor with potential oral bioavailability by targeting two sites in spike protein. *Cell. Dis.* 2022, 8, 88.

10. Peng Sang,+ Yan Shi,+ Bo Huang, Songyi Xue, Timothy Odom, and **Jianfeng Cai**.^{*} Sulfono- γ -AApeptides as helical mimetics: Crystal structures and applications. *Acc. Chem. Res.* 2020, 53, 10, 2425–2442.
9. Peng Sang,+ Zhihong Zhou,+ Yan Shi, Candy Lee, Zaid Amso, David Huang, Timothy Odom, V  n T.B. Nguyen-Tran, Weijun Shen,^{*} and **Jianfeng Cai**.^{*} The Activity of Sulfono- γ -AApeptide Helical Foldamers That Mimic GLP-1. *Sci. Adv.* 2020, 6, 20, eaaz4988.
8. Yan Shi,+ Guangqiang Yin,+ Zhiping Yan, Peng Sang, Minghui Wang, Robert Brzozowski, Prahathees Eswara, Lukasz Wojtas, Youxuan Zheng,^{*} Xiaopeng Li,^{*} and **Jianfeng Cai**.^{*} Helical Sulfono- γ -AApeptides with Aggregation-Induced Emission and Circularly Polarized Luminescence. *J. Am. Chem. Soc.*, 2019, 141, 12697-12706.
7. Peng Sang,+ Min Zhang,+ Yan Shi,+ Chunpu Li, Sami Abdulkadir, Qi Li,^{*} Haitao Ji,^{*} and **Jianfeng Cai**.^{*} Inhibition of β -Catenin/ B-Cell Lymphoma 9 Protein–Protein Interaction Using α -Helix-Mimicking Sulfono- γ -AApeptide Inhibitors. *Proc. Natl. Acad. Sci. U. S. A.*, 2019, 116, 10757-10762.
6. Yan Shi, Sajjan Parag, Rekha Patel, Ashley Lui, Michel Murr, **Jianfeng Cai**^{*}, and Niketa A. Patel^{*}. Stabilization of lncRNA GAS5 by a small molecule and its implications in diabetic adipocytes. *Cell. Chem. Biol.*, 2019, 26, 319-330.
5. Peng Teng, Geoffrey M. Gray, Mengmeng Zheng, Sylvia Singh, Xiaopeng Li, Lukasz Wojtas, Arjan van der Vaart, and **Jianfeng Cai**.^{*} Orthogonal Halogen Bonding Driven 3D Supramolecular Assembly of Right-Handed Synthetic Helical Peptides. *Angew. Chem. Int. Ed.*, 2019, 58, 7778-7782.
4. Fengyu She, Peng Teng, Alfredo Peguero-Tejada, Minghui Wang, Ning Ma, Timothy Odom, Mi Zhou, Erald Gjonaj, Lukasz Wojtas, Arjan van der Vaart, and **Jianfeng Cai**.^{*} De novo Left-Handed Synthetic Peptidomimetic Foldamers, *Angew. Chem. Int. Ed.*, 2018, 9916-9920.
3. Peng Teng, Zheng Niu, Fengyu She, Mi Zhou, Peng Sang, Geoffrey M. Gray, Gaurav Verma, Lukasz Wojtas, Arjan van der Vaart, Shengqian Ma,^{*} and **Jianfeng Cai**.^{*} Hydrogen-Bonding-Driven 3D Supramolecular Assembly of Peptidomimetic Zipper, *J. Am. Chem. Soc.*, 2018, 140, 5661-5665.
2. Peng Teng, Ning Ma, Darrell Cole Cerrato, Fengyu She, Timothy Odom, Xiang Wang, Li-June Ming, Arjan van der Vaart, Lukasz Wojtas, Hai Xu,^{*} and **Jianfeng Cai**.^{*} Right-Handed Helical Foldamers Consisting of de novo D-AApeptides, *J. Am. Chem. Soc.*, 2017, 139, 7363-7369.
1. Yan Shi, Peng Teng, Peng Sang, Fengyu She, Lulu Wei, and **Jianfeng Cai**.^{*} γ -AApeptides: design, structure, and applications. *Acc. Chem. Res.*, 2016, 49, 428-441.

Full List

Work from Independent Career at the University of South Florida:

233. Xue Zhao, Heng Liu, Xiaoyang Lin, Jarais Fontaine, Haiqiang Yang, Virann Welbourn, Chuanhai Cao, and **Jianfeng Cai**.^{*} Sulfonyl- γ -AAs as Turn Templates Inducing β -Sheet Conformation in Macrocyclic Peptides. *J. Med. Chem.* 2026, Accepted.
232. Heng Liu, Xue Zhao, Qiao Qiao, Lukasz Wojtas and **Jianfeng Cai**.^{*} Sulfonyl- γ -AAs as Turn Templates Inducing β -Sheet Conformation in Macrocyclic Peptides. *J. Am. Chem. Soc.* 2026, ASAP.
231. Heng Liu,⁺ Firoz Akhter,⁺ Asma Akhter,⁺ Xue Zhao, Xudong Wang, Haiqiang Yang, Jianyu Chen, Xinyu Xia, Laurent Calcul, Yi He, Chuanhai Cao, Libin Ye, Donghui Zhu,^{*} and **Jianfeng Cai**.^{*} α /Sulfonyl- γ -AApeptide Foldamers Mitigate Alzheimer’s Disease Pathology by Stabilizing Transient Helical Domains in A β . *Nat. Commun.* 2026, ASAP.
230. Peng Sang,^{*} Jiacheng Wei, Yuzhen Qian, Mingshuang Wu, Tingting Song, Wenrui Li, Jinghui Wei, Baoguo Li, Jian Zhang,^{*} **Jianfeng Cai**.^{*} Yan Shi.^{*} Inhibition of Wnt Signaling Using Axin Peptidomimetics through Direct Targeting of β -Catenin. *J. Med. Chem.* 2026, 69, 7, 8583–8598.

229. Chanjuan Dong,+ Sihao Li,+ Xinyu Xia, Timothy H Tran, Satendra Kumar, Huaxuan Yu, Yu Yu Win, Sining Li, Yi He, **Jianfeng Cai**,* Fu-Sen Liang.* Cyclic γ -AApeptide-Based Molecular Glues for RNA m6A Editing. *ACS Chem. Biol.* 2026, 21, 3, 439–445.
228. Liutao Hu, **Jianfeng Cai**,* and Chao Lu.* Structural–Functional Customization of Nanoscale Liposome-in-Liposome Systems: Precision Engineering Methodology and Artificial-Intelligence-Driven Design Prospects. *Research*, 2026, 9, 1168.
227. Mingyu Gong, Zeshi Jiang, Zhitao Yang, Tianxiang Chen, Wanshan Hu, Chunxian Zhou, Jubo Jian, Guilan Quan, Chuanbin Wu, **Jianfeng Cai**,* Chao Lu,* Tingting Peng.* Microneedle technology integrated with diverse therapeutic modalities for hair regrowth in alopecia. *Acta Pharmaceutica Sinica B*. 2026, ASAP.
226. Yunpeng Cui, Zhenghong Chen, Zhikai Li, Bo Li, Shunran Zhang, Xianguang Liu, Mianling Huang, Yinzhi Zhu, Runxu Tang, Hongdi He, Lingang Zhao, Yangyang Wang, Zixiang Yin, Jiang Guo, Songyi Xue, Zhi Chen,* Heng Wang, **Jianfeng Cai**,* Shaodong Zhang,* Xiujun Yu,* and Xiaopeng Li.* Metallo-supramolecular binary necklaces with multiple symmetries. *Sci. China Chem.* 2026, ASAP.
225. Dexin Liu, Andrew Victoria, Alexander Mariscal, Jay Conces, Sandra Mampilly, Yunpeng Cui, Lukasz Wojtas, **Jianfeng Cai**, Wengqi Liu.* Shape-Shifting Tetralactam Macrocycles: Protonation-Activated Convergent Hydrogen Bonding. *Chem. Eur. J.* 2026, 32:e03496.
224. Zainab Fatima, Lihong Chen, Mulan Yin, Yunpeng Cui, **Jianfeng Cai**, Jiandong Chen. MDM2 amplification enables selective PROTAC targeting of tumor cells. *Mol. Cancer Ther.* 2026, ASAP.
223. Zeshi Jiang,+ Wentao Wu,+ Tianxiang Chen, Wanshan Hu, Xinyu Zhang, Chunxian Zhou, Anqi Lu, Yan Yan, Guilan Quan, Chuanbin Wu, Xin Pan, **Jianfeng Cai**,* Chao Lu,* Tingting Peng.* Barrier-Disrupting Microneedle Technology: Overcoming Physical, Physiological, and Pathological Skin Defenses to Enhance Therapeutic Efficacy. *Bioactive Mater.* 2026, 400-429.
222. Peng Sang,*⁺ Jiacheng Wei,⁺ Mingshuang Wu, Qingqing Li, Jian Shang, Baoguo Li, Jian Zhang,* **Jianfeng Cai**,* and Yan Shi.* Disruption of β -catenin/B Cell Lymphoma 9 Protein-Protein Interaction Using Heterogeneous Peptidomimetic Foldamers. *J. Am. Chem. Soc.* 2025, 147, 45, 41819–41829.
221. Lingling Lei, Wangrui Peng, **Jianfeng Cai**,* and Chao Lu.* Combined Ultrasound and Microneedle for Enhanced Transdermal Delivery: Synergistic Mechanisms and Therapeutic Advances. *BIO Integration*, 2025, 6, 31.
220. Makoto Oba, Aya Tanatani, **Jianfeng Cai**, and Peng Teng. Introduction to Foldamers. *Org. Biomol. Chem.* 2025, 23, 9467-9468.
219. Canjia Zhai, Chengkai Xu, Yunpeng Cui, Lukasz Wojtas, **Jianfeng Cai**, Wenqi Liu.* A Synthetic Lectin for Glucuronate. *ACS Central Sci.* 2025, 11, 9, 1753–1761.
218. Bo Huang,+ Emily Gregory-Lott,+ Bingbing X. Li,+ Timothy H. Tran, Sihao Li, Menglin Xue, Shaohui Wang, Anabanadam Asokan, Ning Shen, Xingming Sun, Chuanhai Cao, Xiangshu Xiao,* Gary Daughdrill,* **Jianfeng Cai**.* Discovery of peptidomimetic inhibitors of CREB/CBP by targeting hydrophobic grooves on the surface of the CBP KIX domain. *Acta Pharmaceutica Sinica B*. 2025, 6529-6545.
217. Bo Huang+ Minghui Wang,+ Emily Gregory-Lott,+ Bingbing X. Li,+ Yu Yu Win, Ning Shen, Jianyu Chen, Sihao Li, Chuanhai Cao, Xiangshu Xiao,* Gary W. Daughdrill,* **Jianfeng Cai**.* Recognition of CBP/MLL Interface by Sulfonyl- γ -AApeptides – Beyond Mimicry of MLL. *J. Med. Chem.* 2025, 68, 12272-12283.
216. Yizhan Zhai and **Jianfeng Cai**.* Innovative discovery and mechanistic validation of HyT-PD ligands for selective CDK9-targeted protein degradation. *Acta Pharmaceutica Sinica B*. 2025, 15, 2808-2809.
215. Zejun Xu, Jiaying Chi, Fei Qin, Dongyan Liu, Yecai Lai, Yingxia Bao, Ruizhi Guo, Yiqiu liao, Zhoufan Xie, Jieqiong Jiang, Juyan Liu, **Jianfeng Cai**, Chao Lu, Jiansong Wang, Chuanbin Wu.

- Nanoparticles-incorporated hydrogel microneedle for biomedical applications: fabrication strategies, emerging trends and future prospects. *Asian J. Pharm. Sci.* 2025, 20, 101069.
214. Anna Kharitonova, Rekha S Patel, Brenna Osborne, Meredith Krause-Hauch, Ashley Lui, Gitanjali Vidyarthi, Sihao Li, **Jianfeng Cai**, Niketa A Patel.* NPC86 Increases LncRNA Gas5 in vivo to Improve Insulin Sensitivity and Metabolic Function in Diet-Induced Obese Diabetic Mouse Model. *Int. J. Mol. Sci.* 2025, 26, 8, 3695.
 213. Bo Huang, Sihao Li, Cong Pan, Fangzhou Li, Lukasz Wojtas, Qiao Qiao, Timothy H. Tran, Laurent Calcul, Wenqi Liu, Chenfeng Ke,* **Jianfeng Cai**.* Proline-Based Tripodal Cages with Guest-Adaptive Features for Capturing Hydrophilic and Amphiphilic Fluoride Substances. *Nat. Commun.* 2025, 16, 3226.
 212. Heng Liu, Xue Zhao, Jianyu Chen, Yu Yu Win, and **Jianfeng Cai**.* Unnatural foldamers as inhibitors of A β aggregation via stabilizing A β helix. *Chem. Commun.* 2025, 61, 4586–4594.
 211. Jarais Fontaine and **Jianfeng Cai**.* Recent exploration of γ -AApeptide based antimicrobial peptide mimics as potential therapeutics towards drug-resistant bacteria. *Exploration Drug Sci.* 2025, 3:100888.
 210. Yafeng Wang,+ Xueqing Hu,+ Shriya Pandey,+ Ujjwol Khatri, Tao Shen, Vivek Subbiah, Blaine H. M. Mooers, Ting Chao, Shaohui Wang, Huaxuan Yu, Xingmin Sun, Jie Wu,* and **Jianfeng Cai**.* Targeting oncogenic RET kinase by simultaneously inhibiting kinase activity and degrading the protein. *J. Med. Chem.* 2025, 68, 1, 81–94
 209. Haohui Ye and **Jianfeng Cai**.* Cationic Catalysts as a New Strategy to Catalyze *N*-Carboxyanhydrides Polymerization. *ACS Central Sci.* 2025, 11, 3, 376-378.
 208. Jinhua Xie,+ Shahedul Islam,+ Le Wang,* Xiaojing Zheng, Mengsheng Xu, Xiqi Su, Shaohua Huang, Logan Suits, Guang Yang, Praathees Eswara, **Jianfeng Cai**, Li-June Ming.* A tale of two old drugs tetracycline and salicylic acid with new perspectives—Coordination chemistry of their Co(II) and Ni(II) complexes, redox activity of Cu(II) complex, and molecular interactions. *J. Inorg. Biochem.* 2025, 262, 112757.
 207. Zhikai Li, Yu Yan, Zhi Chen, Runxu Tang, Ningxu Han, Runlin Han, Fang Fang, Jichun Jiang, Lei Hua, Xiujun Yu, Ming Wang, **Jianfeng Cai**, Haiyang Li, Heng Wang* and Xiaopeng Li.* Multi-Dimensional Decryption of Metallopolymer at Single-Chain Level. *CCS Chem.* 2025, 7, 2381-2393.
 206. Wei Jiang, Jiayan Chen, Haifeng Wang, Aiqi Xue, Xinyang Zhang, Jichi Guan, Lulu Wei, **Jianfeng Cai**,* Yong Hu,* and Dan Liu.* Design, Synthesis and Pharmacological Evaluation of Novel 4-Phenoxyquinoline Derivatives as VEGFR2 Kinase Inhibitors for Tumor Treatment. *Chem. Res. Chin. Univ.* 2025, 41, 66-78.
 205. Canjia Zhai, Ethan Cross Zulueta, Alexander Mariscal, Chengkai Xu, Yunpeng Cui, Xudong Wang, Huang Wu, Carson Doan, Lukasz Wojtas, Haixin Zhang, **Jianfeng Cai**, Libin Ye, Kun Wang, Wenqi Liu.* From Small Changes to Big Gains: Pyridinium-Based Tetralactam Macrocycle for Enhanced Sugar Recognition in Water. *Chem. Sci.* 2024, 15, 19588-19598.
 204. Ruixuan Gao, Menglin Xue, Ning Shen, Xue Zhao, Justin C. Zhang, Chuanhai Cao, and **Jianfeng Cai**.* Development of low-toxicity antimicrobial polycarbonates bearing lysine residues. *Chem. Eur. J.* 2024, e202402302.
 203. Ying Feng, YongChao Zhu, Tian Chen, Pengcheng Li, Bingjie Liu, **Jianfeng Cai**,* Wenjie Liang,* and Hai Xu.* Green and efficient preparation and application of weakly crystalline TiO₂ with high catalytic activity. *New J. Chem.*, 2024, 48, 515-519.
 202. Yuwei Zheng, Jiaying Chi, Jiayu Ou, Ling Jiang, Liqing Wang, Rui Luo, Yilang Yan, Zejun Xu, Tingting Peng, **Jianfeng Cai**, Chuanbin Wu, Peng Teng,* Guilan Quan,* Chao Lu.* Imidazole-Rich, Four-Armed Host-Defense Peptidomimetics as Promising Narrow-Spectrum Antibacterial Agents and Adjuvants against *Pseudomonas Aeruginosa* Infections. *Adv. Healthcare Mater.*, 2024, 2400664.

201. Tengyue Jian,+ Minghui Wang,+ Jeevapani Hettige, Yuhao Li, Lei Wang, Ruixuan Gao, Wenchao Yang, Renyu Zheng, Shengliang Zhong, Marcel D. Baer, Aleksandr Noy, James. J. De Yoreo, **Jianfeng Cai**,* Chun-Long Chen.* Self-assembling and pore-forming peptoids as novel antimicrobial biomaterials. *ACS Nano*, 2024, 18, 34, 23077–23089.
200. Shannon J. Ho, Dale Chaput, Rachel G. Sinkey, Amanda H. Garces, Erika P. New, Maja Okuka, Peng Sang, Sefa Arlier, Nihan Semerci, Thora S. Steffensen, Thomas J. Rutherford, Angel E. Alsina, **Jianfeng Cai**, Matthew L. Anderson, Ronald R. Magness, Vladimir N. Uversky, Derek A. T. Cummings & John C. M. Tsibris. *Cell Commun. Signaling* 2024, 22, 221.
199. Xue Zhao, Heng Liu, Justin C. Zhang, **Jianfeng Cai**. * Helical Sulfonyl- γ -AApeptides in the Inhibition of HIV-1 Fusion and HIF-1 α Signaling. *RSC Med. Chem.* 2024, 15, 1418-1423.
198. Heng Liu, Yunpeng Cui, Xue Zhao, Lulu Wei, Xudong Wang, Ning Shen, Timothy Odom, Xuming Li, William Lawless, Kanchana Karunaratne, Martin Muschol, Wayne Guida, Chuanhai Cao, Libin Ye, and **Jianfeng Cai**. * Helical sulfonyl- γ -AApeptides modulating A β oligomerization and cytotoxicity by recognizing A β helix. *Proc. Natl. Acad. Sci. U. S. A.* 2024, 121, 6, e2311733121.
197. Menglin Xue,+ Soumyadeep Chakraborty,+ Ruixuan Gao,+ Shaohui Wang, Meng Gu, Ning Shen, Lulu Wei, Chuanhai Cao, Xingmin Sun,* and **Jianfeng Cai**. * Antimicrobial Guanidinylate Polycarbonates Show Oral In Vivo Efficacy Against Clostridioides difficile. *Adv. Healthcare Mater.* 2024, 2303295.
196. Prakash Jadhav,+ Bo Huang,+ Jerzy Osipiuk,+ Xiaoming Zhang, Haozhou Tan, Christine Tesar, Michael Endres, Robert Jedrzejczak, Bin Tan, Xufang Deng, Andrzej Joachimiak,* **Jianfeng Cai**,* Jun Wang.* Structure-based design of SARS-CoV-2 papain-like protease inhibitors. *Eur. J. Med. Chem.* 2024, 264, 116011.
195. Xiaomin Guo,+ Xiaokang Miao,+ Yingying An, Tiantian Yan, Yue Jia, Bochuan Deng, **Jianfeng Cai**, Wenle Yang, Wangsheng Sun,* Rui Wang,* Junqiu Xie.* Novel antimicrobial peptides modified with fluorinated sulfono- γ -AA having high stability and targeting multidrug-resistant bacteria infections. *Eur. J. Med. Chem.* 2024, 116001.
194. Wenhao Wang, Chao Lu,* , Zhengwei Huang,* , Lei Shu, **Jianfeng Cai**, Chuanbin Wu and Xin Pan. A Bibliometric Study on Nanomedicines as Melanoma Therapeutics: Clinical Translation is Urgent. *Oncol. Adv.* 2023, 1, 25-30.
193. Zhanpeng Zhang, Shuai Lu,* Xiujun Yu, Lei Hua, Weiguo Wang, Menglin Xue, **Jianfeng Cai**, Heng Wang,* Xiaopeng Li. Construction of Metallo-Helicoids via Intermolecular Coordination with High Antimicrobial Activity. *Chem. Commun.* 2023, 59, 13022.
192. Yuqing Tong,+ Meng Gu,+ Xingyu Luo, Haifeng Qi, Wei Jiang, Yu Deng, Lulu Wei, Jun Liu, Yin Ding,* **Jianfeng Cai**,* Yong Hu.* An engineered nanoplatfrom cascade to relieve extracellular acidity and enhance resistance-free chemotherapy. *J. Control. Release.* 2023, 363, 562-573.
191. Wei Jiang,+ Sami Abdulkadir,+ Xue Zhao,+ Peng Sang, Anastasia Tomatsidou, Xiujun Zhang, Yu Chen, Laurent Calcul, Xingmin Sun, Feng Cheng,* Yong Hu,* **Jianfeng Cai**. * Inhibition of Hypoxia-Inducible Transcription Factor (HIF-1 α) Signaling with Sulfonyl- γ -AApeptide Helices. *J. Am. Chem. Soc.* 2023, 145, 36, 20009-20020.
190. Songyi Xue,+ Wei Xu,+* Lei Wang, Ling Xu, Laurent Calcul, Peng Teng, Lu Lu, Shibo Jiang,* and **Jianfeng Cai**. * Rational Design of Sulfonyl- γ -AApeptides as Highly Potent HIV-1 Fusion Inhibitors with Broad-spectrum Activity. *J. Med. Chem.* 2023, 66, 18, 13319–13331.
189. Ali Azmy,+ Xue Zhao,+ Giasemi K. Angeli, Claire Welton, Parth Raval, Lukasz Wojtas, Nouridine Zibouche, G. N. Manjunatha Reddy, Pantelis N. Trikalitis, **Jianfeng Cai**, and Ioannis Spanopoulos.* One Year Water Stable and Porous Bi(III) Halide Semiconductor with Broad Spectrum Antibacterial Performance. *ACS App. Mater. Inter.* 2023, 15, 36, 42717–42729.
188. Peng Sang* and **Jianfeng Cai**. * Unnatural Helical Peptidic Foldamers as Protein Segment Mimics. *Chem. Soc. Rev.* 2023, 52, 4843-4877.

187. Seid Yimer Abate,+ Ziqi Yang,+ Surabhi Jha, Jada Emodogo, Guorong Ma, Zhongliang Ouyang, Shafi Muhammad, Nihar Pradhan, Xiaodan Gu, Derek Patton, Dawen Li, **Jianfeng Cai**,* and Qilin Dai.* Promoting large area slot-die coated perovskite solar cells performance and reproducibility by acid-based sulfono- γ -AApeptide. *ACS Appl. Mater. Inter.* 2023, 15, 36, 42717–42729.
186. Hongtao Kong,+ Shangshang Qin,+ Dachao Yan, Boyuan Shen, Tingting Zhang, Meng Wang, Sen Li, Maxwell Ampomah-Wireko, Mengmeng Bai, En Zhang,* **Jianfeng Cai**.* Development of aromatic-linked diamino acid antimicrobial peptide mimics with low hemolytic toxicity and excellent activity against methicillin-resistant Staphylococcus aureus (MRSA). *J. Med. Chem.* 2023, 66, 12, 7756–7771.
185. Songyi Xue,+ Wei Xu,+* Lei Wang, Xinling Wang, Qianyu Duan, Laurent Calcul, Shaohui Wang, Wenqi Liu, Xingmin Sun, Lu Lu,* Shibo Jiang,* and **Jianfeng Cai**.* An HR2-mimicking sulfonyl- γ -AApeptide is a potent pan-coronavirus fusion inhibitor with strong blood-brain barrier permeability, long half-life and promising oral bioavailability. *ACS Central. Sci.* 2023, 9 1046-1058.
184. Diego Alem, Xinrui Yang, Francisca Beato, Bhaswati Sarcar, Alexandra F. Tassielli, Ruifan Dai, Tara L. Hogenson, Margaret A. Park, Kun Jiang, **Jianfeng Cai**, Yu Yuan, Martin E. Fernandez-Zapico, Aik Choon Tan, Jason B. Fleming, Hao Xie.* Translational relevance of SOS1 targeting for KRAS-mutant colorectal cancer. *Mol. Carcinogenesis*, 2023, 62, 1025-1037.
183. Yafeng Wang, Menglin Xue, Ruixuan Gao, Soumyadeep Chakraborty, Shaohui Wang, Xue Zhao, Meng Gu, Chuanhai Cao, Xinmin Sun, **Jianfeng Cai**.* Short lipidated dendrimeric γ -AApeptides as new antimicrobial peptidomimetics. *Int. J. Mol. Sci.*, 2023, 24, 7, 6407.
182. Lei Wang,+ Chunlong Ma,+ Michael Dominic Sacco,+ Songyi Xue, Mentalla Mahmoud, Laurent Calcul, Yu Chen,* Jun Wang,* and **Jianfeng Cai**.* Development of the Safe and Broad-Spectrum Aldehyde and Ketoamide Mpro inhibitors Derived from the Constrained α , γ -AA Peptide Scaffold. *Chem. Eur. J.* 2023, e202300476.
181. Meng Gu,+ Ying Yu,+ Menglin Xue, Jianxiong Jiang,* **Jianfeng Cai**.* The discovery of cyclic γ -AApeptides as the promising ligands targeting EP2. *Bioorg. Med. Chem. Lett.* 2023, 87, 129255.
180. Peng Teng,* Haodong Shao, Bo Huang, Junqiu Xie, Sunliang Cui,* Kairong Wang,* **Jianfeng Cai**.* Small Molecular Mimetics of Antimicrobial Peptides as a Promising Therapy to Combat Bacterial Resistance. *J. Med. Chem.* 2023, 66, 4, 2211–2234.
179. Oksana Fihurka, Yanhong Wang, Yuzhu Hong, Xiaoyang Lin, Ning Shen, Haiqiang Yang, Breanna Brown, Marcus Mommer, Tarek Zieneldien, Yitong Li, Janice Kim, Minghua Li, **Jianfeng Cai**, Qingyu Zhou,* Chuanhai Cao.* Multi-targeting intranasal nanoformulation as a therapeutic for Alzheimer's disease. *Biomolecules*, 2023, 13, 232.
178. Ruixuan Gao, Xuming Li, Menglin Xue, Ning Shen, Minghui Wang, Jingyao Zhang, Chuanhai Cao, and **Jianfeng Cai**.* Development of Lipidated Polycarbonates with Broad-Spectrum Antimicrobial Activity. *Biomater. Sci.* 2023, 11, 1840-1852
177. Xiaoqian Feng, Dongyi Xiana, Jintao Fu, Rui Luo, Wenhao Wang, Yuwei Zheng, Qing He, Zhan Ouyang, Shaobin Fang, Wancong Zhang, Daojun Liu, Shijie Tang, Guilan Quan, **Jianfeng Cai**, Chuanbin Wu, Chao Lu,* Xin Pan.* Four-armed host-defense peptidomimetics-augmented vanadium carbide MXene-based microneedle array for efficient photo-excited bacteria-killing. *Chem. Eng. J.* 2023, 456, 141121.
176. Rekha S. Patel, Ashley Lui, Charles Hudson, Lauren Moss, Robert Sparks, Shannon E. Hill, Yan Shi, **Jianfeng Cai**, Laura J. Blair, Paula Bickford, Niketa A. Patel. Small molecule targeting long noncoding RNA GAS5 administered intranasally improves neuronal insulin signaling and decreases neuroinflammation in an aged mouse model. *Sci. Rep.* 2023, 13, 373.
175. Seid Yimer Abate,+ Ziqi Yang,+ Surabhi Jha, Guorong Ma, Zhongliang Ouyang, Haixin Zhang, Shafi Muhammad, Nihar Pradhan, Xiaodan Gu, Derek Patton, Kun Wang, Dawen Li, **Jianfeng Cai**,* Qilin Dai.* Room Temperature Slot-Die Coated Perovskite Layer Modified with sulfonyl- γ -AApeptide for High Performance Perovskite Solar Devices. *Chem. Eng. J.* 2023, 457, 141199.

174. Li Zhou,+ In Ho Jeong,+ Songyi Xue, Menglin Xue, Lei Wang, Sihao Li, Ruochuan Liu, Geon Ho Jeong, Xiaoyu Wang, **Jianfeng Cai**,* Jun Yin,* Bo Huang.* Inhibition of the ubiquitin transfer cascade by peptidomimetic foldamer mimicking E2 N-terminal-helix. *J. Med. Chem.* 2023, 66, 1, 491-502.
173. Xueying Ge, Fangchao Jiang, Minghui Wang, Meng Chen, Yiming Li, Joshua Phipps, **Jianfeng Cai**, Jin Xie, Jane Ong, Viktor Dubovoy, James G. Masters, Long Pan,* Shengqian Ma.* Naringin@Metal-Organic Framework as a Multifunctional Bio-platform. *ACS Applied Mater. Interface.* 2023, 15, 1, 677-683.
172. Pengcheng Li, Xiaohuan Chen, Cui Guo, Huijing Zou, Ziyi Chen, Bingjie Liu, Wenjie Liang,* **Jianfeng Cai**,* Hai Xu.* A Logic Fluorescent Chemosensor Based on Eu³⁺ Functionalized Cd-MOFs for Sensing Fe³⁺ and Cu²⁺ Synchronously. *Eur. J. Inorg. Chem.* 2023, 26, e2022005
171. Xiaomin Guo, Tiantian Yan, Jing Rao, Yingying An, Xin Yue, Xiaokang Miao, Rui Wang, Wangsheng Sun,* **Jianfeng Cai**,* and Junqiu Xie.* Novel Feleucin-K3-derived peptides modified with sulfono- γ -AA building blocks targeting *Pseudomonas aeruginosa* and MRSA infections. *J. Med. Chem.* 2023, 66, 2, 1254-1272.
170. Yan Shi,+ Candy Lee,+ Peng Sang, Zaid Amso, David Huang, Weixia Zhong, Meng Gu, Lulu Wei, V n T. B. Nguyen-Tran, Jingyao Zhang, Weijun Shen,* **Jianfeng Cai**,* α /Sulfono- γ -AA peptide hybrids agonist of GLP-1R with prolonged action both in vitro and in vivo. *Acta Pharmaceutica Sinica B*, 2023, 66, 2, 1648-1659.
169. Yujia Bian, Diego Alem, Francisca Beato, Tara L. Hogenson, Xinrui Yang, Kun Jiang, **Jianfeng Cai**, Wen Wee Ma, Martin Fernandez-Zapico, Aik Choon Tan, Nicholas J. Lawrence, Jason B. Fleming, Yu Yuan*, Hao Xie.* Development of SOS1 inhibitor-based degraders to target KRAS-mutant colorectal cancer. *J. Med. Chem.* 2022, 65, 16432-16450.
168. Songyi Xue,+ Xinling Wang,+ Lei Wang, Wei Xu, Shuai Xia, Lujia Sun, Shaohui Wang, Ning Shen, Ziqi Yang, Bo Huang, Sihao Li, Chuanhai Cao, Laurent Calcul, Xingmin Sun, Lu Lu,* **Jianfeng Cai**,* and Shibo Jiang.* A novel cyclic $\hat{I}^3$ -AApeptide-based long-acting pan-coronavirus fusion inhibitor with potential oral bioavailability by targeting two sites in spike protein. *Cell. Dis.* 2022, 8, 88.
167. Jie Zhong, Yuegui Guo, Shaoyong Lu, Kun Song, Ying Wang, Li Feng, Zhen Zheng, Qiufen Zhang, Jiacheng Wei, Peng Sang, Yan Shi, **Jianfeng Cai**, Guoqiang Chen, Chen-Ying Liu,* Xiuyan Yang,* and Jian Zhang.* Rational design of a sensitivity-enhanced tracer for discovering efficient APC-Asef inhibitors. *Nat. Commun.* 2022, 13(1):4961.
166. Songyi Xue,+ Lei Wang,+ **Jianfeng Cai**,* Sulfono- $\hat{I}^3$ -AApeptides as protein helical domain mimetics to manipulate the angiogenesis. *ChemBioChem* 2022, 23, e202200298.
165. Xing Zhang,+ Minghui Wang,+ Xiaodi Zhu, Yan Peng, Tiwei Fu, Chang-Hua Hu, **Jianfeng Cai**,* Guojian Liao.* Development of lipo- γ -AA peptides as potent antifungal agents. *J. Med. Chem.* 2022, 65, 8029-8039.
164. Chao Lu, Feng Li, Liming Lin, Jiaying Chi, Hui Wang, Minqun Du, Disang Feng, Liqing Wang, Rui Luo, Hangping Chen, Guilan Quan, **Jianfeng Cai**, Xin Pan,* Chuanbin Wu.* Guanidinium-Rich Lipopeptide Functionalized Bacteria-Absorbing Sponge as an Effective Trap-and-Kill System for the Elimination of Focal Bacterial Infection. *Acta Biomaterialia*, 2022, 148, 106-118.
163. Peng Sang, Yan Shi, Lulu Wei, and **Jianfeng Cai**,* Helical Sulfono- γ -AApeptides with Predictable Functions in Protein Recognition. *RSC Chem. Biol.* 2022, 3, 805-814.
162. Yanhong Wang, Yuzhu Hong, Jiyu Yan, Breanna Brown, Xiaoyang Lin, Xiaolin Zhang, Ning Shen, Minghua Li, **Jianfeng Cai**, Marcia Gordon, David Morgan, Qingyu Zhou,* and Chuanhai Cao.* Low-Dose Delta-9-Tetrahydrocannabinol as Beneficial Treatment for Aged APP/PS1 Mice. *Int. J. Mol. Sci.* 2022, 23, 5, 2757.

161. Wei Jiang,+ Lulu Wei,+ Bing Chen, Xingyu Luo, Peipei Xu,* **Jianfeng Cai**,* and Yong Hu.* Platinum Prodrug Nanoparticles Inhibiting Tumor Recurrence and Metastasis by Concurrent Chemoradiotherapy. *J. Nano. Biotechnol.* 2022, 20:129.
160. Jintao Fu,Ting Liu, Xiaoqian Feng, Yixian Zhou, Minglong Chen, Wenhao Wang, Yiting Zhao, Chao Lu, Guilan Quan, **Jianfeng Cai**, Xin Pan*, Chuanbin Wu. Perfect Pair: Stabilized Black Phosphorous Nanosheets Engineering with Antimicrobial Peptides for Robust Multi-Drug Resistant Bacteria Eradication. *Adv Healthc Mater.* 2022, e2101846.
159. Peng Teng* and **Jianfeng Cai**.* Using Proteomimetics to Switch Angiogenic Signaling. *Acta Pharmaceutica Sinica B*, 2022, 1534-1535.
158. Bo Huang,+ Li Zhou,+ Ruochuan Liu,+ Lei Wang, Songyi Xue, Yan Shi, Geon Ho Jeong, In Ho Jeong, Sihao Li, Jun Yin,* **Jianfeng Cai**.* Activation of E6AP/UBE3A-Mediated Protein Ubiquitination and Degradation Pathways by a Cyclic γ -AApeptide. *J. Med. Chem.* 2022, 65, 2497-2506.
157. Sami Abdulkadir,+ Chunpu Li,+ Wei Jiang,+ Xue Zhao, Peng Sang, Lulu Wei, Yong Hu,* Qi Li,* and **Jianfeng Cai**.* Modulating Angiogenesis by Proteomimetics of Vascular Endothelial Growth Factor. *J. Am. Chem. Soc.*, 2022, 144, 1, 270–281.
156. Lulu Wei and **Jianfeng Cai**.* Novel Peptides and Peptidomimetics in Drug Discovery. *Acta Pharmaceutica Sinica B*, 2021, 2606-2608.
155. Maochao Zheng, Huanchang Lina, Wancong Zhang, Shijie Tang,* Daojun Liu,* and **Jianfeng Cai**.* Poly(L-ornithine)-grafted zinc phthalocyanines as dual-functional antimicrobial agents with intrinsic membrane damage and photothermal ablation capacity. *ACS Infect. Dis.* 2021, 7, 2917–2929.
154. Peng Sang,+ Hongxiang Zeng,+ Candy Lee,+ Yan Shi, Minghui Wang, Cong Pan, Lulu Wei, Chenglong Huang, Mingjun Wu, Weijun Shen,* Xi Li,* and **Jianfeng Cai**.* α /sulfono- γ -AApeptide Hybrid Analogues of Glucagon with Enhanced Stability and Prolonged in vivo Activity. *J. Med. Chem.* 2021, 64, 13893–13901.
153. Mengmeng Zheng,+ Chunpu Li,+ Mi Zhou,+ Ru Jia, Gang Cai, Fengyu She, Lulu Wei, Shaohui Wang, Jie Yu, Dingyan Wang, Laurent Calcul, Xingmin Sun, Xiaomin Luo, Feng Cheng, Qi Li,* Yan Wang,* and **Jianfeng Cai**.* Discovery of Cyclic Peptidomimetic Ligands Targeting the Extracellular Domain of EGFR. *J. Med. Chem.* 2021, 64, 11219-11228.
152. Wei Jiang, Xingyu Luo, Lulu Wei, Shanmei Yuan, **Jianfeng Cai**,* Xiqun Jiang,* and Yong Hu.* The Sustainability of Energy Conversion Inhibition for Tumor Ferroptosis Therapy and Chemotherapy. *Small*, 2021, 17, 2102695.
151. Mengmeng Zheng,+ Chunpu Li,+ Mi Zhou,+ Ru Jia, Fengyu She, Lulu Wei, Feng Cheng, Qi Li, * **Jianfeng Cai*** and Yan Wang.* Peptidomimetic-based antibody surrogate for HER2. *Acta Pharmaceutica Sinica B*, 2021, 2645-2654.
150. Song Qing, Han Zhifen, Xinnan Wu, Yan Wang, Lihong Zhou, Liu Yang, Ningning Liu, Hua Sui, **Jianfeng Cai**, Qing Ji,* Li Qi.* β -arrestin1 promotes colorectal cancer metastasis through GSK-3 β / β -catenin signaling-mediated epithelial-to-mesenchymal transition. *Frontiers Cell Develop. Biol.* 2021, 9, 650067.
149. Yan Shi, Peng Sang and **Jianfeng Cai**.*. Discovery of α -Helix-Mimicking Sulfono- γ -AApeptides as p53–MDM2 inhibitors. *Future. Med. Chem.* 2021, 13, 12, 1021-1023.
148. Minghui Wang,+ Xiaoqian Feng,+ Ruixuan Gao, Peng Sang, Xin Pan, Lulu Wei, Chao Lu,* Chuanbin Wu,* **Jianfeng Cai**.* Modular Design of Membrane Active Antibiotics: From Macromolecular Antimicrobials to Small Scorpion-like Peptidomimetics. *J. Med. Chem.* 2021, 9894-9905.
147. Peng Teng,*+ Mengmeng Zheng,+ Darrell Cole Cerrato, Yan Shi, Mi Zhou, Songyi Xue, Wei Jiang, Lukasz Wojtas, Li-June Ming, Yong Hu,* **Jianfeng Cai**.* The Folding Propensity of α /Sulfono- γ -AA Peptidic Foldamers with Both Left- and Right-Handedness. *Commun. Chem.* 2021, 4, 58.

146. Lulu Wei,+ Ruixuan Gao,+ Minghui Wang,+ Yafeng Wang, Yan Shi, Meng Gu and **Jianfeng Cai**.^{*} Dimeric Lipo- α /sulfono- γ -AA Hybrid Peptides as Broad-Spectrum Antibiotic Agents. *Biomater. Sci.* 2021, 6, 3207-3217.
145. Yin Shi, Xiaoqian Feng, Jing Wang, Limin Lin, Biyuan Wu, Guilin Zhou, Feiyuan Yu, Qian Xu, Daojun Liu, Guilan Quan, Chao Lu,* Xin Pan,* **Jianfeng Cai**,* Chuanbin Wu.* Virus-Inspired Surface-Nanoengineered Antimicrobial Liposome: A potential System to Simultaneously Achieve High Activity and Selectivity. *Bioactive Mat.* 2021, 6, 3207–3217.
144. Maochao Zheng, Miao Pan, Wancong Zhang, Huanchang Lin, Shenlang Wu, Chao Lu,* Shijie Tang,* Daojun Liu,* **Jianfeng Cai**.^{*} Poly(α -L-lysine)-based nanomaterials for versatile biomedical applications: Current advances and perspectives. *Bioactive Mat.* 2021, 6, 1878-1909.
143. **Jianfeng Cai*** and Ruihui Liu.* Introduction to Antibacterial Biomaterials. *Biomater. Sci.* 2020, 8, 6812-6813.
142. Qing Song, Liu Yang, Zhifen Han, Xinnan Wu, Ruixiao Li, Lihong Zhou, Ningning Liu, Hua Sui, **Jianfeng Cai**, Yan Wang, Qing Ji,* and Qi Li.* Tanshinone IIA Inhibits Epithelial-to-Mesenchymal Transition Through Hindering β -Arrestin1 Mediated β -Catenin Signaling Pathway in Colorectal Cancer. *Front. Pharmacol.* 2020, 11, 586616.
141. Yan Shi,+ Peng Sang,+ Junhao Lu,+ Pirada Higbee,+ Lihong Chen, Leixiang Yang, Timothy Odom, Gary Daughdrill,* Jiandong Chen,* and **Jianfeng Cai**.^{*} Rational Design of Right-Handed Heterogeneous Peptidomimetics as Inhibitors of Protein–Protein Interactions. *J. Med. Chem.* 2020, 63, 21, 13187–13196.
140. Minghui Wang, Ruixuan Gao, Mengmeng Zheng, Peng Sang, Chunpu Li, En Zhang, Qi Li, and **Jianfeng Cai**.^{*} Development of Bis-Cyclic Imidazolidine-4-one Derivatives as Potent Antibacterial Agents. *J. Med. Chem.* 2020, 63, 24, 15591–15602.
139. Peng Sang,+ Yan Shi,+ Bo Huang, Songyi Xue, Timothy Odom, and **Jianfeng Cai**.^{*} Sulfono- γ -AApeptides as helical mimetics: Crystal structures and applications. *Acc. Chem. Res.* 2020, 53, 10, 2425–2442.
138. Runzhe Song, Yue Wang, Minghui Wang, Ruixuan Gao, Teng Yang, Song Yang, Cai-Guang Yang, Yongsheng Jin, Siyuan Zou, **Jianfeng Cai**,* Renhua Fan,* Qiuqin He.* Design and synthesis of novel desfluoroquinolone-aminopyrimidine hybrids as potent anti-MRSA agents with low hERG activity. *Bioorg. Chem.* 2020, 104176.
137. Miao Pan,+ Chao Lu,+ Maochao Zheng, Wen Zhou, Fuling Song, Weidong Chen, Fen Yao, Daojun Liu,* and **Jianfeng Cai**.^{*} Unnatural Amino Acid-Based Star-Shaped Poly(L-ornithine)s as Emerging Long-Term and Biofilm-Disrupting Antimicrobial Peptides to Treat Pseudomonas aeruginosa Infected Burn Wounds. *Adv. Healthcare. Mat.*, 2020, 9, 19, 2000647.
136. Ge Song, Haiqiang Yang, Ning Shen, Phillip Pham, Breanna Brown, Xiaoyang Lin, Yuzhu Hong, Paul Sinu, **Jianfeng Cai**, Xiaopeng Li, Michael Leon, Marcia Gordon, David Morgan, Sai Zhang, and Chuanhai Cao. An Immunomodulatory Therapeutic Vaccine Targeting Oligomeric Amyloid Beta. *J. Alz. Dis.*, 2020, 77, 4, 1639-1653.
135. Peng Sang,+ Yan Shi,+ Pirada Higbee, Minghui Wang, Sami Abdulkadir, Junhao Lu, Gary W. Daughdrill,* Jiandong Chen,* and **Jianfeng Cai**.^{*} Rational Design and Synthesis of Right-Handed D-Sulfono- γ -AApeptide Helical Foldamers as Potent Inhibitors of Protein-Protein Interactions. *J. Org. Chem.* 2020, 85, 10552–10560.
134. Chengming Fu, Tian Chen, Tile Xiao, Yuecai Song, Timothy Odom, Wenjie Liang,* **Jianfeng Cai**.^{*} Hai Xu.* Formaldehyde Gas Adsorption in High-Capacity Silver-Nanoparticle-Loaded ZIF-8 and UiO-66 Frameworks. *ChemistrySelect*, 2020, 5987-5992.
133. Hao Liu, Shijie Yi, Yunfeng Wu, Han Wu, Jinrong Zhou, Wenjie Liang,* **Jianfeng Cai**.^{*} and Hai Xu.* An efficient Co-N/C electrocatalyst for oxygen reduction facilely prepared by tuning cobalt species content. *Int. J. Hydro. Energy* 2020, 16105-16113.

132. Minghui Wang, Ruixuan Gao, Peng Sang, Tomothy Odom, Mengmeng Zheng, Yan Shi, Hai Xu, Chuanhai Cao and **Jianfeng Cai**.^{*} Dimeric γ -AApeptides with Potent and Selective Antibacterial Activity. *Front. Chem.* 2020, 8, 441.
131. Jing Wang,+ Chao Lu,+ Yin Shi, Xiaoqian Feng, Biyuan Wu, Guilin Zhou, Guilan Quan, Xin Pan,^{*} **Jianfeng Cai**,^{*} and Chuanbin Wu.^{*} Structural Superiority of Guanidinium-Rich Four-Armed Copolypeptides: Role of Multiple Peptide-Membrane Interactions in Enhancing Bacterial Membrane Perturbation and Permeability. *ACS. Appl. Mater. Inter.* 2020, 12, 18363-18374.
130. Yan Shi,+ Peng Sang,+ Guangqiang Yin, Ruixuan Gao Xiao Liang, Robert Brzozowski, Prahathees Eswara, Timothy Odom, Youxuan Zheng,^{*} Xiaopeng Li,^{*} and **Jianfeng Cai**.^{*} Aggregation-Induced Emissive and Circularly Polarized Homogeneous Sulfono- γ -AApeptide Foldamers. *Adv. Opt. Mater.* 2020, 8, 1902122.
129. Peng Sang,+ Zhihong Zhou,+ Yan Shi, Candy Lee, Zaid Amso, David Huang, Timothy Odom, V  n T.B. Nguyen-Tran, Weijun Shen,^{*} and **Jianfeng Cai**.^{*} The Activity of Sulfono- γ -AApeptide Helical Foldamers That Mimic GLP-1. *Sci. Adv.* 2020, 6, 20, eaaz4988.
128. Minghui Wang, Timothy Odom, and **Jianfeng Cai**.^{*} Challenges in the development of next-generation antibiotics: Opportunities of small molecules mimicking mode of action of host-defense peptides. *Exp. Opin. Ther. Pat.* 2020, 303-305.
127. De-Yun Cui, Hong-Tao Kong, Yi Yang, **Jianfeng Cai**, Bo-Yuan Shen, Da-chao Yan, Xiu-Juan Zhang, Ying-Long Qu, Meng-Meng Bai, En Zhang.^{*} Asymmetric synthesis of linezolid thiazolidine-2-thione derivatives via CS₂ mediated decarboxylation cyclization. *Tetrahedron Lett.* 2020, 151847.
126. Qing Ji, Lihong Zhou, Hua Sui, Liu Yang, Xinnan Wu, Qing Song, Ru Jia, Ruixiao Li, Jian Sun, Ziyuan Wang, Ningning Liu, Yuanyuan Feng, Xiaoting Sun, Gang Cai, Yu Feng, **Jianfeng Cai**, Yihai Cao, Guoxiang Cai, yan wang, and Qi Li.^{*} Primary tumors release ITGBL1-rich extracellular vesicles to promote distal metastatic tumor growth through fibroblast-niche formation. *Nat. Commun.* 2020, 11, 1211.
125. Peng Sang,+ Yan Shi,+ Junhao Lu,+ Lihong Chen, Leixiang Yang, Wade Borchers, Sami Abdulkadir, Qi Li,^{*} Gary Daughdrill,^{*} Jiandong Chen,^{*} and **Jianfeng Cai**.^{*} α -Helix-Mimicking Sulfono- γ -AApeptide Inhibitors for p53-MDM2/MDMX Protein-Protein Interactions. *J. Med. Chem.* 2020, 63, 3, 975-986.
124. Mi Zhou,+ Mengmeng Zheng,+ and **Jianfeng Cai**.^{*} Small molecules with membrane-active antibacterial activity. *ACS. Appl. Mater. Inter.* 2020, 12, 19, 21292-21299.
123. Olapeju Bolarinwa, Chunpu Li, Nawal Khadka, Qi Li, Yan Wang, Jianjun Pan,^{*} and **Jianfeng Cai**.^{*} γ -AApeptides-based Small Molecule Ligands That Disaggregate Human Islet Amyloid Polypeptide. *Sci. Rep.*, 2020, 2020, 10, 95.
122. Sylvia Singh, Minghui Wang, Ruixuan Gao, Peng Teng, Timothy Odom, En Zhang, Hai Xu, and **Jianfeng Cai**.^{*} Lipidated α /Sulfono- α -AA heterogeneous peptides as antimicrobial agents for MRSA. *Bioorg. Med. Chem.*, 2020, 28, 115241.
121. Lulu Wei, Minghui Wang, Ruixuan Gao, Rojin Fatirkhorani and **Jianfeng Cai**.^{*} Antibacterial activity of lipo- α /sulfono- γ -AA hybrid peptides. *Eur. J. Med. Chem.*, 2020, 186, 11901.
120. Ma Su, Yan Shi, Minghui Wang, Ruixuan Gao, Jianfeng Wu, Hai Xu, Chuanwu Xi,^{*} and **Jianfeng Cai**.^{*} The Activity of Small Urea- γ -AApeptides Toward Gram-Positive Bacteria. *ChemMedChem*, 2019, 14, 1963-1967.
119. Ma Su, Minghui Wang, Yuzhu Hong, Alekhya Nimmagadda, Ning Shen, Yan Shi, Ruixuan Gao, En Zhang, Chuanhai Cao,^{*} and **Jianfeng Cai**.^{*} Polymyxin Derivatives as Broad-Spectrum Antibiotic Agents. *Chem. Commun.*, 2019, 55, 13104-13107.
118. Wei Jiang, Chao Zhang, Arsalan Ahmed, Yunlei Zhao, Yu Deng, Yin Ding,^{*} **Jianfeng Cai**,^{*} and Yong Hu.^{*} H₂O₂-Sensitive Upconversion Nanocluster Bomb for Tri-Mode Imaging-Guided Photodynamic Therapy in Deep Tumor Tissue. *Adv. Healthcare Mat.*, 2019, 8, 1900972.

117. Chao Lu, Guilan Quan, Ma Su, Alekhya Nimmagadda, Weidong Chen, Miao Pan, Peng Teng, Feiyuan Yu, Xi Liu, Ling Jiang, Wenyi Du, Wei Hu, Fen Yao, Xin Pan, Chuanbin Wu,* Daojun Liu,* and **Jianfeng Cai**.* Molecular Architecture and Charging Effects Enhance the in Vitro and in Vivo Performance of Multi-Arm Antimicrobial Agents Based on Star-Shaped Poly(L-lysine). *Advanced Therapeutics*, 2019, 1900147.
116. Heng Wang,+ Chung-Hao Liu,+ Kun Wang, Minghui Wang, Hao Yu, Sneha Kandapal, Robert Brzozowski, Bingqian Xu, Ming Wang, Shuai Lu, Xin-Qi Hao, Prahathes Eswara, Mu-Ping Nieh,* **Jianfeng Cai**,* and Xiaopeng Li.* Assembling Pentatopic Terpyridine Ligand with Three Types of Coordination Moieties into Giant Supramolecular Hexagonal Prism: Synthesis, Self-Assembly, Characterization, and Antimicrobial Study. *J. Am. Chem. Soc.*, 2019, 141, 16108-16116.
115. Yong Liang, Xiang Wang, Siqi Zhao, Piao He, Ting Luo, Jinzhi Jiang, Wenjie Liang,* **Jianfeng Cai**,* and Hai Xu.* A New Photoresponsive Bis (Crown Ether) for Extraction of Metal Ions. *ChemistrySelect*, 2019, 4, 10316-10319.
114. Yan Shi,+ Guangqiang Yin,+ Zhiping Yan, Peng Sang, Minghui Wang, Robert Brzozowski, Prahathes Eswara, Lukasz Wojtas, Youxuan Zheng,* Xiaopeng Li,* and **Jianfeng Cai**.* Helical Sulfono- γ -AApeptides with Aggregation-Induced Emission and Circularly Polarized Luminescence. *J. Am. Chem. Soc.*, 2019, 141, 12697-12706.
113. Wenchao Chu, Yi Yang, **Jianfeng Cai**, Hongtao Kong, Mengmeng Bai, Xiangjing Fu, Shangshang Qin, En Zhang.* Synthesis and Bioactivities of New Membrane-active Agents with Aromatic Linker: High Selectivity and Broad-Spectrum Antibacterial Activity. *ACS. Infect. Dis.*, 2019, 13, 1535-1545.
112. Yan Shi and **Jianfeng Cai**.* Discovery of a Macrocyclic γ -AApeptide binding to lncRNA GAS5 and its therapeutic implication in Type 2 Diabetes. *Future. Med. Chem.*, 2019, 11, 2233-2235.
111. Simon S. Terzyan, Tao Shen, Xuan Liu, Qingling Huang, Peng Teng, Mi Zhou, Frank Hilberg, **Jianfeng Cai**, Blaine H.M. Mooers, and Jie Wu.* Structural basis of resistance of mutant RET protein tyrosine kinase to its inhibitors nintedanib and vandetanib. *J. Biol. Chem.*, 2019, 294, 10428-10437.
110. Ning Shen, Ge Song, Haiqiang Yang, Xiaoyang Lin, Breanna Brown, Yuzhu Hong, **Jianfeng Cai**, Chuanhai Cao.* Identifying the pathological domain of alpha- synuclein as a therapeutic for Parkinson's disease. *Int. J. Mol. Sci.*, 2019, 20, 2338.
109. Peng Sang, Min Zhang, Yan Shi, Chunpu Li, Sami Abdulkadir, Qi Li,* Haitao Ji,* and **Jianfeng Cai**.* Inhibition of β -Catenin/ B-Cell Lymphoma 9 Protein-Protein Interaction Using α -Helix-Mimicking Sulfono- γ -AApeptide Inhibitors. *Proc. Natl. Acad. Sci. U. S. A.*, 2019, 116, 10757-10762.
108. Peng Teng, Geoffrey M. Gray, Mengmeng Zheng, Sylvia Singh, Xiaopeng Li, Lukasz Wojtas, Arjan van der Vaart, and **Jianfeng Cai**.* Orthogonal Halogen Bonding Driven 3D Supramolecular Assembly of Right-Handed Synthetic Helical Peptides. *Angew. Chem. Int. Ed.*, 2019, 58, 7778-7782.
107. Qing Yin, Tao Han, Bin Fang, Guolin Zhang, Chao Zhang, Evan R. Roberts, Victoria Izumi, Mengmeng Zheng, Shulong Jiang, Xiu Yin, Minjung Kim, **Jianfeng Cai**, Eric B. Haura, John M. Koomen, Keiran S. M. Smalley and Lixin Wan.* K27-linked Ubiquitination of BRAF by ITCH Engages Cytokine Response to Maintain MEK-ERK Signaling. *Nat. Commun.*, 2019, 10, 1870.
106. Chu Wenchao, Yi Yang, Shangshang Qin, **Jianfeng Cai**, Mengmeng Bai, Kong Hongtao and En Zhang.* Low-toxicity Amphiphilic Molecules linked by an Aromatic Nucleus Show Broad-spectrum Antibacterial Activity and Low Drug Resistance. *Chem. Commun.*, 2019, 55, 4307-4310.
105. Hao Yan, Mi Zhou, Umesh Bhattarai, Yabin Song, Mengmeng Zheng, **Jianfeng Cai**,* Fu-Sen Liang.* Cyclic peptidomimetics as inhibitors for miR-155 biogenesis. *Mol. Pharm.*, 2019, 914-920.
104. Zhong Peng, Shaohui Wang, Mussie Gide, Duolong Zhu, Chunhui Li, **Jianfeng Cai**, Xingmin Sun.* A novel bacteriophage lysin-human defensin fusion protein is effective in treatment of Clostridioides difficile infection in mice. *Frontiers in Microbiology*, 2019, 9, 3234.

103. Ting Luo, Hao Liu, Yong Liang, Jun Tang, Jinrong Zhou, Wenjie Liang,* **Jianfeng Cai**,* and Hai Xu.* A Comparison of Drug Delivery Systems of Zr-Based MOFs and Halloysite Nanotubes: Evaluation of β -Estradiol Encapsulation. *ChemistrySelect*, 2019, 4, 8925-8929.
102. Yan Shi, Sajjan Parag, Rekha Patel, Ashley Lui, Michel Murr, **Jianfeng Cai***, and Niketa A. Patel*. Stabilization of lncRNA GAS5 by a small molecule and its implications in diabetic adipocytes. *Cell. Chem. Biol.*, 2019, 26, 319-330.
101. Alekhya Nimmagadda, Yan Shi and **Jianfeng Cai**.* γ -AApeptides as a new strategy for therapeutic development. *Curr. Med. Chem.*, 2019, 26, 1-16.
100. Mussie Gide, Alekhya Nimmagadda, Ma Su, Minghui Wang, Peng Teng, Chunpu Li, Ruixuan Gao, Hai Xu, Qi Li,* **Jianfeng Cai***. Nano-Sized Lipidated Dendrimers as Potent and Broad Spectrum Antibacterial Agents, *Macromol. Rapid Commun.*, 2018, 1800622.
99. Olapeju Bolarinwa, Meng Zhang, Erin Mulry, Min Lu* and **Jianfeng Cai**.* Sulfono- γ -AA modified peptides that inhibit HIV-1 fusion, *Org. Biomol. Chem.*, 2018, 7878-7882.
98. Jisong Hua, Peng Teng, Yingying Zou, Chao Zhang, Xujie Shen, **Jianfeng Cai**,* and Yong Hu.* Small Antimicrobial Agents encapsulated in poly(epsilon-caprolactone)-poly(ethylene glycol) nanoparticles for treatment of S. aureus -infected wounds, *J. Nanopar. Res.*, 2018, 20:270.
97. Fengyu She, Peng Teng, Alfredo Peguero-Tejada, Minghui Wang, Ning Ma, Timothy Odom, Mi Zhou, Erald Gjonaj, Lukasz Wojtas, Arjan van der Vaart, and **Jianfeng Cai**.* De novo Left-Handed Synthetic Peptidomimetic Foldamers, *Angew. Chem. Int. Ed.*, 2018, 9916-9920.
96. Olapeju Bolarinwa and **Jianfeng Cai**.* Developments with investigating descriptors for antimicrobial AApeptides and their derivatives., *Exp. Opin. Drug. Discov.*, 2018 , 727-739.
95. Yong Liang, Jun Tang, Xiang Wang, Siqi Zhao, Ting Luo, Cijun Shuai,* Jinzhi Jiang, **Jianfeng Cai***, and Hai Xu.* Using bispyrene fluorescence probe for determining the multiple states of organogel. *Chemistryselect*, 2018, 5361-5363.
94. Sylvia Singh, Alekhya Nimmagadda, Ma Su, Minghui Wang, Peng Teng, and **Jianfeng Cai**.* Lipidated α/α -AA heterogeneous peptides as antimicrobial agents, *Eur. J. Med. Chem.*, 2018 , 398-405.
93. Peng Teng, Chunhui Li, Zhong Peng, Anne Marie Vanderschouw, Alekhya Nimmagadda, Ma Su, Yaqiong Li, Xingmin Sun,* and **Jianfeng Cai**.* Facilely Accessible Quinoline Derivatives as Potent Antibacterial Agents, *Bioorg. Med. Chem.*, 2018 , 3573-3579.
92. Chunhui Li, Peng Teng, Zhong Peng, Peng Sang, Xingmin Sun,* and **Jianfeng Cai**.* Bis-Cyclic-Guanidine as a Novel Class of Compounds Potent Against Clostridium Difficile, *ChemMedChem*, 2018 , 1414-1420.
91. Heng Wang, Xiaomin Qian, Kun Wang, Ma Su, Wei-Wei Haoyang, Xin Jiang, Robert Brzozowski, Ming Wang, Xiang Gao, Yiming Li, Bingqian Xu, Prahathees Eswara, Xin-Qi Hao, Weitao Gong,* Jun-Li Hou,* **Jianfeng Cai***, Xiaopeng Li.* Supramolecular Kandinsky Circles with High Antibacterial Activity, *Nat. Commun.*, 2018, 9, 1815.
90. Peng Teng, Zheng Niu, Fengyu She, Mi Zhou, Peng Sang, Geoffrey M. Gray, Gaurav Verma, Lukasz Wojtas, Arjan van der Vaart, Shengqian Ma,* and **Jianfeng Cai**.* Hydrogen-Bonding-Driven 3D Supramolecular Assembly of Peptidomimetic Zipper, *J. Am. Chem. Soc.*, 2018, 140, 5661-5665.
89. Youhong Niu,* Minghui Wang, Yafei Cao, Alekhya Nimmagadda, Jianxing Hu, Yanfen Wu, **Jianfeng Cai***, and Xin-Shan Ye.* Rational Design of Dimeric Lysine N-Alkylamides as Potent and Broad-Spectrum Antibacterial Agents, *J. Med. Chem.*, 2018, 61, 2865-2874.
88. Xiaojun Sun, Yuan Ren, Steven Gunawan, Peng Teng, Zhengming Chen, Harshani Lawrence, **Jianfeng Cai**, Nicholas Lawrence, and Jie Wu.* Selective inhibition of leukemia-associated SHP2E69K mutant by the allosteric SHP2 inhibitor SHP099, *Leukemia*, 2018, 32, 1246-1249.
87. Yan Shi, Sridevi Challa, Peng Sang, Fengyu She, Chunpu Li, Geoffrey M. Gray, Alekhya Nimmagadda, Peng Teng, Timothy Odom, Yan Wang, Arjan van der Vaart, Qi Li,* and **Jianfeng Cai**.* One-Bead-

- Two-Compound Thioether Bridged Macrocyclic γ -AApeptide Screening Library against EphA2, *J. Med. Chem.*, 2017, 60, 9290-9298.
86. Peng Teng, Alekhya Nimmagadda, Ma Su, Yuzhu Hong, Ning Shen, Chunpu Li, Ling-Yu Tsai, Jessica Cao, Qi Li,* and **Jianfeng Cai**.* Novel Bis-Cyclic Guanidines as Potent Membrane-Active Antibacterial Agents with Therapeutic Potential, *Chem. Commun.*, 2017, 53, 11948-11951.
 85. Chao Zhang, Xiao cheng, Mengkun Chen, Jie Sheng, Jing Ren, Zhongying Jiang,* **Jianfeng Cai**,* and Yong Hu.* Fluorescence guided photothermal/photodynamic ablation of tumours using pH-responsive chlorin e6-conjugated gold nanorods, *Colloids Surfaces B: Biointerfaces*, 2017, 160, 345-354.
 84. Ma Su, Donglin Xia, Peng Teng, Alekhya Nimmagadda, Chao Zhang, Timothy Odom, Annie Cao, Yong Hu, and **Jianfeng Cai**.* Membrane-Active Hydantoin Derivatives as Antibiotic Agents, *J. Med. Chem.*, 2017, 60, 8456-8465.
 83. Hua Sui, Jihui Zhao, Lihong Zhou, Haotian Wen, Wanli Deng, Chunpu Li, Qing Ji, Xuan Liu, Yuanyuan Feng, Ni Chai, Qibo Zhang, **Jianfeng Cai**, Qi Li.* Tanshinone IIA inhibits β -catenin/VEGF-mediated angiogenesis by targeting TGF- β 1 in normoxic and HIF-1 α in hypoxic microenvironments in human colorectal cancer, *Cancer Lett.*, 2017, 403, 86-97.
 82. Zhe Zhang, Heng Wang, Xu Wang, Yiming Li, Bo Song, Olapeju Bolarinwa, R. Alexander Reese, Tong Zhang, Xu-Qing Wang, **Jianfeng Cai**, Bingqian Xu, Ming Wang,* Changlin Liu,* Hai-Bo Yang, and Xiaopeng Li.* Super Snowflakes: Step-Wise Self-Assembly and Dynamic Exchange of Rhombus Star-Shaped Supramolecules, *J. Am. Chem. Soc.*, 2017, 139, 8174-8185.
 81. Peng Teng, Ning Ma, Darrell Cole Cerrato, Fengyu She, Timothy Odom, Xiang Wang, Li-June Ming, Arjan van der Vaart, Lukasz Wojtas, Hai Xu,* and **Jianfeng Cai**.* Right-Handed Helical Foldamers Consisting of de novo D-AApeptides, *J. Am. Chem. Soc.*, 2017, 139, 7363-7369.
 80. Jianjun Pan,* Prasana K. Sahoo, Annalisa Dalzini, Zahra Hayati, Chinta M. Aryal, Peng Teng, **Jianfeng Cai**, Humberto Rodriguez Gutierrez, Likai Song.* Membrane Disruption Mechanism of a Prion Peptide (106-126) Investigated by Atomic Force Microscopy, Raman and Electron Paramagnetic Resonance Spectroscopy, *J. Phys. Chem. B.*, 2017, 121, 5058-5071.
 79. Hai Xu,* Siqi Zhao, Xiang Xiong, Jinzhi Jiang, Wei Xu, Daoben Zhu, Yi Zhang, Wenjie Liang, **Jianfeng Cai**.* Atomic Force Microscope characterization of self-assembly behaviors of cyclo[8] pyrrole on solid substrates, *Chem. Phys. Lett.*, 2017, 647, 151.
 78. Nawal K Khadka; Peng Teng, **Jianfeng Cai**, and Jianjun Pan.* Modulation of Lipid Membrane Structural and Mechanical Properties by a Peptidomimetic Derived from Reduced Amide Scaffold. *Biochim. Biophys. Acta.*, 2017, 1859, 734-744.
 77. Olapeju Bolarinwa, Alekhya Nimmagadda, Ma Su, and **Jianfeng Cai**.* Structure and Function of AApeptides. *Biochemistry*, 2017, 445-457.
 76. Alekhya Nimmagadda, Xuan Liu, Peng Teng, Ma Su, Yaqiong Li, Qiao Qiao, Nawal K Khadka, Xiaoting Sun, Jianjun Pan, Hai Xu,* Qi Li,* and **Jianfeng Cai**.* Polycarbonates with Potent and Selective Antimicrobial Activity toward Gram-Positive Bacteria. *Biomacromolecules*, 2017, 18, 87-95.
 75. Peng Sang, Yan Shi, Peng Teng, Annie Cao, Hai Xu, Qi Li, and **Jianfeng Cai**.* Antimicrobial AApeptides. *Curr. Top. Med. Chem.*, 2017, 17, 1266-1279.
 74. Peng Teng, Da Huo, Alekhya Nimmagadda, Jianfeng Wu, Fengyu She, Ma Su, Xiaoyang Lin, Jiyu Yan, Annie Cao, Chuanwu Xi,* Yong Hu,* and **Jianfeng Cai**.* Small antimicrobial agents based on acylated reduced amide scaffold. *J. Med. Chem.*, 2016, 59, 7877-7887.
 73. Fengyu She, Alekhya Nimmagadda, Peng Teng, Ma Su, Xiaobing Zuo, and **Jianfeng Cai**.* Helical 1:1 α /sulfono- γ -AA heterogeneous peptides with antibacterial activity. *Biomacromolecules*, 2016, 17, 1854-1859.
 72. Fengyu She, Olapeju Oyesiku, Peiguang Zhou, Shiming Zhuang, David W. Koenig, and **Jianfeng Cai**.* The development of Antimicrobial γ -AApeptides. *Future Med. Chem.*, 2016, 8, 1101.

71. Chian Sing Ho, Nawal K. Khakda, Fengyu She, **Jianfeng Cai**, and Jianjun Pan.* Influenza M2 Transmembrane Domain Senses Membrane Heterogeneity and Enhances Membrane Curvature. *Langmuir*, 2016, 32, 6730-6738.
70. Pavanjeet Kaur, Yaqiong Li, **Jianfeng Cai**,* and Likai Song.* Selective Membrane Disruption Mechanism of an Antibacterial γ -AApeptide Defined by EPR Spectroscopy. *Biophys. J.*, 2016, 110, 1789-1799.
69. Peng Teng, Yan Shi, Peng Sang, and **Jianfeng Cai**.* γ -AApeptides as a new class of peptidomimetics. *Chem. Eur. J.*, 2016, 22, 2-11.
68. Yan Shi, Peng Teng, Peng Sang, Fengyu She, Lulu Wei, and **Jianfeng Cai**.* γ -AApeptides: design, structure, and applications. *Acc. Chem. Res.*, 2016, 49, 428-441.
67. Hai Xu,* Siqi Zhao, Yang Ren, Wei Xu, Daoben Zhu, Jinzhi Jiang and **Jianfeng Cai**.* Primary Investigation of optical limiting performance of Cyclo [8] pyrrole with wide optical limiting window. *RSC Advances*, 2016, 6, 21067-21071.
66. Chian Sing Ho, Nawal K Khakda, Fengyu She, **Jianfeng Cai**, and Jianjun Pan.* Polyglutamine Aggregates Impair Lipid Membrane Integrity and Enhance Lipid Membrane Rigidity. *Biochim. Biophys. Acta.*, 2016, 1858, 661-670.
65. Yan Wang, Frankie Costanza, Alekhya Nimmagadda, Daqian Song, **Jianfeng Cai**,* and Qi Li.* PEGpoly (amino acid)s/MicroRNA complex nanoparticles effectively arrest the growth and metastasis of colorectal cancer, *J. Biomed. Nanotechnol.*, 2016, 12, 1510-1519.
64. Xiaoyang Lin, Ge Bai, Kyle Sutherland, Frankie Costanza, Kurt Breitenkamp, Kevin Sill, **Jianfeng Cai**,* and Chuanhai Cao.* Polymer-Encapsulated A β Peptide Fragments as an Oligomeric-Specific Vaccine for Alzheimer's disease" *J. Biomed. Nanotechnol.*, 2016, 12, 1421-1430.
63. Haifan Wu, Jinzhi Jiang, Hai Xu, Qi Li, **Jianfeng Cai**.* RGD mimetic γ -AApeptides and methods of use us 20140004039 a1: a patent evaluation. *Expert Opin. Ther. Pat.*, 2016, 26, 131-137.
62. Fan Chao, Lu Chen, Qingling Huang, Tao Shen, Eric A. Welsh, Jamie K. Teer, **Jianfeng Cai**, W. Douglas Cress, and Jie Wu.* Overexpression of major CDKN3 transcripts is associated with poor survival in lung adenocarcinoma. *Br. J. Cancer*, 2015, 113, 1735-1743.
61. Hua Sui, Hanchen Xu, Qing Ji, Xuan Liu, Lihong Zhou, Haiyan Song, Xiqiu Zhou, Yangxian Xu, Zhesheng Chen, **Jianfeng Cai**, Guang Ji, Qi Li.* 5-hydroxytryptamine receptor (5-HT1DR) promotes colorectal cancermetastasis by regulating Axin1/ β -catenin/MMP-7 signaling pathway. *Oncotarget*. 2015, 25975-25987.
60. Haifan Wu, Qiao Qiao, Peng Teng, Yaogang Hu, Dimitrios Antoniadis, Xiaobing Zuo, and **Jianfeng Cai**.* A new class of heterogeneous helical peptidomimetics. *Org. Lett.*, 2015, 17 (14), 3524-3527.
59. Yaqiong Li, Haifan Wu, Peng Teng, Ge Bai, Xiaoyang Lin, Xiaobing Zuo, Chuanhai Cao, **Jianfeng Cai**.* Helical antimicrobial sulfono- γ -AApeptides. *J. Med. Chem.*, 2015, 58, 4802-4811.
58. Yuxia Hao, Ge Bai, Junping Wang, Longfeng Zhao, Kyle Sutherland, **Jianfeng Cai** and Chuanhai Cao.* Identifiable biomarker and treatment development using HIV-1 long term non-progressor sera. *BMC Immunol*, 2015, 16:25.
57. Shruti Padhee, Yaqiong Li, **Jianfeng Cai**.* Activity of lipo-cyclic γ -AApeptides against biofilms of staphylococcus epidermidis and pseudomonas aeruginosa. *Bioorg. Med. Chem. Lett.*, 2015, 25, 2565-2569.
56. Haifan Wu, Fengyu She, Wen-Yang Gao, Austin Prince, Yaqiong Li, Lulu Wei, Allison Mercer, Lukasz Wojtas, Shengqian Ma, and **Jianfeng Cai**.* The Synthesis of Head-to-Tail Cyclic SulfonoyAApeptides. *Org. Biomol. Chem.*, 2015, 13, 672-676.

55. Haifan Wu, Qiao Qiao, Yaogang Hu, Peng Teng, Wenyang Gao, Xiaobing Zuo, Lukasz Wojtas, Randy W. Larsen, Shengqian Ma, and **Jianfeng Cai**.* Sulfono- γ -AApeptides as a new class of unnatural helical foldamer. *Chem. Eur. J.*, 2015, 21, 2501-2507.
54. Qing Ji, Xuan Liu, Zhifen Han, Lihong Zhou, Hua Sui, Linlin Yan, Haili Jiang, Jianlin Ren, **Jianfeng Cai**, and Qi Li.* Resveratrol suppresses epithelial-to-mesenchymal transition in colorectal cancer through TGF- β 1/Smads signaling pathway mediated Snail/E-cadherin expression. *BMC Cancer*, 2015, 15:97.
53. Kenneth E. Ugen, Xiaoyang Lin, Ge Bai, Zhanhua Liang, **Jianfeng Cai**, Kunyun Li, Shijie Song, Chuanhai Cao* and Juan Sanchez-Ramos. Evaluation of an alpha synuclein sensitized dendritic cell based vaccine in a transgenic mouse model of Parkinson's disease. *Hum. Vaccin. Immunother.*, 2015, 11, 922-930.
52. Peng Teng, Haifan Wu, Lili Lin and **Jianfeng Cai**.* Antimicrobial γ -AApeptides (WO2013112548)-a patent evaluation. *Expert Opin. Ther. Pat.*, 2015, 25, 111-118.
51. Yaogang Hu, Ni Cheng, Haifan Wu, Samuel Kang, Richard D. Ye,* and **Jianfeng Cai**.* Design, synthesis and characterization of fMLF-mimicking AApeptides. *ChemBioChem*, 2014, 15, 2420-2426.
50. Yaqiong Li, Christina Smith, Haifan Wu, Peng Teng, Yan Shi, Shruti Padhee, Torey Jones, Anh-My Nguyen, Chuanhai Cao, Hang Yin,* and **Jianfeng Cai**.* Short antimicrobial lipo- α/γ -AA hybrid peptides. *ChemBioChem*, 2014, 2074-2280.
49. Peng Teng, Xiaolei Zhang, Haifan Wu, Qiao Qiao, Said M Sebti* and **Jianfeng Cai**.* Identification of novel inhibitors that disrupt STAT3/DNA interaction from γ -AApeptide OBOC combinatorial library. *Chem. Commun.* 2014, 50, 8739 - 8742.
48. Xiaoyang Lin, Ge Bai, Linda Lin, Hengyi Wu, **Jianfeng Cai**, Kenneth E Ugen*, Chuanhai Cao*. Vaccination induced changes in pro-inflammatory cytokine levels as an early putative biomarker for cognitive improvement in a transgenic mouse model for Alzheimer disease. *Hum. Vaccin. Immunother.* 2014, 10(7), 2024-2031.
47. Chuanhai Cao*, Yaqiong Li, Hui Liu, Ge Bai, Xiaoyang Lin, Kyle Sutherland, Jonathan Myal, Neel Nabar, **Jianfeng Cai***. The potential therapeutic effects of THC on Alzheimer's disease. *J. Alz. Dis.* 2014, 973-984.
46. Yan Wang, Frankie Costanza, Haifan Wu, Daqian Song, **Jianfeng Cai*** and Qi Li*. PEG-poly (amino acid)s-encapsulated Tanshinone IIA as potential therapeutics for the treatment of hepatoma. *J. Mat. Chem. B.* 2014, 3115-3112.
45. Yan Wang, Daqian Song, Frankie Costanza, Huirong Zhu, Zhongze Fan,* **Jianfeng Cai*** and Qi Li.* Targeted Delivery of Tanshinone IIA-conjugated mPEG-PLGA-PLL-cRGD Nanoparticles to Hepatocellular Carcinoma. *J. Biomed. Nanotechnol.* 2014, 3244-3252.
44. Wen-Yang Gao, Yao Chen, Youhong Niu, Kia Williams, Lindsay Cash, Pastor Perez, Lukasz Wojtas , **Jianfeng Cai**, Yu-Sheng Chen and Shengqian Ma*. Crystal engineering of an nbo topology MOF for chemical fixation of CO₂ under ambient conditions. *Angew Chem. Int. Ed.*, 2014, 53, 2615-2619.
43. Shruti Padhee, Christina Smith, Haifan Wu, Yaqiong Li, Namitha Manoj, Qiao Qiao, Zoya Khan, Chuanhai Cao, Hang Yin,* and **Jianfeng Cai**.* The development of antimicrobial γ -AApeptides that suppress pro-inflammatory immune responses. *ChemBioChem*, 2014, 688-694.
42. Haifan Wu, Peng Teng and **Jianfeng Cai**.* Quick access to multiple classes of peptidomimetics from common γ -AApeptide building blocks. *Eur. J. Org.*, 2014, 1760-1765.
41. Yaqiong Li, Christina Smith, Haifan Wu, Shruti Padhee, Namitha Manoj, Joseph Cardiello, Qiao Qiao, Chuanhai Cao, Hang Yin,* and **Jianfeng Cai**.* Lipidated cyclic γ -AApeptides display both antimicrobial and anti-inflammatory activity. *ACS Chem. Biol.*, 2014, 9, 211-217.

40. Haifan Wu, Yaqiong Li, Ge Bai, Youhong Niu, Qiao Qiao, Jeremiah Tipton, Chuanhai Cao,* **Jianfeng Cai**.* γ -AApeptide-based small-molecule ligands that inhibit A β aggregation. *Chem. Commun.*, 2014, 50, 5206-208.
39. Frankie Costanza, Shruti Padhee, Haifan Wu, Yan Wang, Jesse Revenis, Chuanhai Cao, Qi Li* and **Jianfeng Cai**.* Investigation of antimicrobial PEG-poly(amino acid)s. *RSC Advances*, 2014, 4, 20892095.
38. Rongsheng E. Wang,* Yin Zhang, Ling Tian, Weibo Cai* and **Jianfeng Cai**. Antibody-Based Imaging of HER-2: Moving into the Clinic. *Curr. Mol. Med.*, 2013, 13, 1523-1537.
37. Yaqiong Li, Haifan Wu, Youhong Niu, Yaogang Hu, Qi Li, Chuanhai Cao, **Jianfeng Cai**.* Development of RNA Aptamer-Based Therapeutic Agents. *Curr. Med. Chem.*, 2013, 20, 3655-3663.
36. Haifan Wu, Peng Teng, Youhong Niu, Qi Li, **Jianfeng Cai**.* Polymyxin derivatives: a patent evaluation (WO2012168820). *Expert Opin. Ther. Pat.*, 2013, 1075-81.
35. Youhong Niu, Haifan Wu, Yaqiong Li, Yaogang Hu, Shruti Padhee, Qi Li, Chuanhai Cao and **Jianfeng Cai**.* AApeptides as a new class of antimicrobial agents. *Org. Biomol. Chem.* 2013, 11, 4283-4290.
34. Long Zhang, Qing Ji, Xuan Liu, Xingzhu Chen, Zhao Hua Chen, Yanyan Qiu, Jian Sun, **Jianfeng Cai**, Huirong Zhu, and Qi Li. Norcantharidin inhibits tumor angiogenesis via blocking VEGFR2/MEK/ERK signaling pathways. *Cancer Sci.*, 2013, 104, 604-610.
33. Neel R. Nabar, Fang Yuan, Xiaoyang Lin, Li Wang, Ge Bai, Jonathan Mayl, Yaqiong Li, Shu-Feng Zhou, Jinhuan Wang, **Jianfeng Cai**, Chuanhai Cao*. Cell Therapy: A Safe and Efficacious Therapeutic Treatment for Alzheimer's Disease in APP+PS1 Mice. *PLoS One*, 2012, 7, 12, e49468.
32. Youhong Niu, Haifan Wu, Rongfu Huang, Qiao Qiao, Frankie Costanza, Xi-Sen Wang, Yaogang Hu, Mohamad Nassir Amin, Anh-My Nguyen, James Zhang, Edward Haller, Shengqian Ma, Xiao Li, and **Jianfeng Cai**.* Nanorods formed from a new class of peptidomimetics. *Macromolecules*, 2012, 45, 7350–7355.
31. Yaogang Hu, Mohamad Nassir Amin, Shruti Padhee, Rongsheng E. Wang, Qiao Qiao, Ge Bai, Yaqiong Li, Archana Mathew, Chuanhai Cao, and **Jianfeng Cai**.* Lipidated Peptidomimetics with Improved Antimicrobial Activity. *ACS Med. Chem. Lett.* 2012, 55, 4003-4009.
30. Youhong Niu, Rongsheng E. Wang*, Haifan Wu, **Jianfeng Cai**.* Recent development of small antimicrobial peptidomimetics. *Future Med. Chem.* 2012, 4, 14, 1853-1862.
29. Haifan Wu, Mohamad Nassir Amin, Youhong Niu, Qiao Qiao, Nassier Harfouch, Abdelfattah Nimer, **Jianfeng Cai**.* Solid Phase Synthesis of γ -AApeptides Using a Novel Submonomeric Approach. *Org. Lett.* 2012, 14, 3446-3449.
28. Yunan Yang, Youhong Niu, Hao Hong, Haifan Wu, Yin Zhang, Jonathan W. Engle, Todd E. Barnhart, **Jianfeng Cai***, and Weibo Cai*. Radiolabeled γ -AApeptides: A New Class of Tracers for Positron Emission Tomography. *Chem. Commun.* 2012, 48, 7850-7852.
27. Haifan Wu, Youhong Niu, Shruti Padhee, Rongsheng E Wang, Yaqiong Li, Qiao Qiao, Ge Bai, Chuanhai Cao, and **Jianfeng Cai**.* Design and synthesis of unprecedented cyclic γ -AApeptides for antimicrobial development. *Chem. Sci.*, 2012, 3, 2570-2575.
26. Zhongqiu Luo, Jialin Li, Neel R. Nabar, Xiaoyang Lin, Ge Bai, **Jianfeng Cai**, Shu-Feng Zhou, Chuanhai Cao*, Jinhuan Wang*. Efficacy of a Therapeutic Vaccine Using Mutated β -amyloid Sensitized Dendritic Cells in Alzheimer's Mice. *J. Neuroimmune Pharmacol.*, 2012, 7, 640-645.
25. Wen-Yang Gao , Youhong Niu , Yao Chen , Lukasz Wojtas , **Jianfeng Cai**, Yu-Sheng Chen and Shengqian Ma*. Porous Metal-Organic Framework Based on a Macrocyclic Tetracarboxylate Ligand Exhibiting Selective CO₂ Uptake. *CrystEngComm*, 2012, 14, 6115-6117.
24. Youhong Niu, Shruti Padhee, Haifan Wu, Ge Bai, Qiao Qiao, Yaogang Hu, Lacey Harrington, Whitney N. Burda, Lindsey N. Shaw, Chuanhai Cao, and **Jianfeng Cai**.* Lipo- γ -AApeptides as a new class of potent and broad-spectrum antimicrobial agents. *J. Med. Chem.* 2012, 55(8), 4003–4009.

23. Chuanhai Cao*, David A. Loewenstein, Xiaoyang Lin, Chi Zhang, Li Wang, Ranjan Duara, Yougui Wu, Alessandra Giannini, Ge Bai, **Jianfeng Cai**, Maria Greig, Elizabeth Schofield, Raj Ashok, Brent Small, Huntington Potter and Gary W. Arendash*. High Blood Caffeine Levels in MCI Linked to Lack of Progression to Dementia. *J. Alz. Dis.* 2012, 30, 559-572.
22. Youhong Niu, Ge Bai, Haifan Wu, Rongsheng E. Wang, Qiao Qiao, Shruti Padhee, Robert Buzzeo, Chuanhai Cao*, and **Jianfeng Cai***. Cellular translocation of a γ -AApeptide mimetic of Tat peptide. *Mol. Pharmaceutics*. 2012, 9(5), 1529–1534
21. Ge Bai, Shruti Padhee, Youhong Niu, Rongsheng E. Wang, Robert Buzzeo, Chuanhai Cao*, and **Jianfeng Cai***. Cellular uptake of an α -AApeptide. *Org. Biomol. Chem.* 2012, 10 (6), 1149 - 1153.
20. Rongsheng E. Wang,* Frankie Costanza, Youhong Niu, Haifan Wu, Yaogang Hu, Whitney Hang, Yiqun Sun, **Jianfeng Cai***. Development of self-immolative dendrimers for drug delivery and sensing. *J. Control. Release*. 2012, 159, 154-163.
19. Rongsheng E. Wang, Youhong Niu, Haifan Wu, Yaogang Hu, **Jianfeng Cai***. Development of NGRBased Anti-Cancer Agents for Targeted Therapeutics and Imaging. *Anticancer Agents Med. Chem.* 2012, 12 (1), 76-86.
18. Youhong Niu, Shruti Padhee, Haifan Wu, Ge Bai, Lacey Harrington, Whitney N. Burda, Lindsey N. Shaw, Chuanhai Cao, and **Jianfeng Cai***. Identification of γ -AApeptides with potent and broadspectrum antimicrobial activity. *Chem. Commun.* 2011, 47 (44), 12197 - 12199.
17. Rongsheng E. Wang, Yin Zhang, **Jianfeng Cai**, Weibo Cai, Ting Gao*. Aptamer-Based Fluorescent Biosensors. *Curr. Med. Chem.* 2011, 18, 4175-4184.
16. Rongsheng E. Wang,* Haifan Wu, Youhong Niu, and **Jianfeng Cai***. Improving the Stability of Aptamers by Chemical Modification. *Curr. Med. Chem.* 2011, 18, 4126-4138.
15. Rongsheng E. Wang, Youhong Niu, Haifan Wu, Mohamad Nassir Amin, and **Jianfeng Cai***. Development of NGR peptide-based agents for tumor imaging. *Am. J. Nucl. Med. Mol. Imaging.*, 2011, 1(1), 36-46.
14. Shruti Padhee, Yaogang Hu, Youhong Niu, Ge Bai, Haifan Wu, Frankie Costanza, Leigh West, Lacey Harrington, Lindsey N. Shaw, Chuanhai Cao, and **Jianfeng Cai***. Non-Hemolytic α -AApeptides as Antimicrobial Peptidomimetics. *Chem. Commun.* 2011, 47 (34), 9729 - 9731
13. Youhong Niu, Alisha “Jonesy” Jones, Haifan Wu, Gabriele Varani,* and **Jianfeng Cai***. γ -AApeptides bind to RNA by mimicking RNA-binding proteins. *Org. Biomol. Chem.*, 2011, 9 (19), 6604 - 6609.
12. Youhong Niu, Yaogang Hu, Xiaolong Li, Jiandong Chen, and **Jianfeng Cai***. Gamma-AApeptides: Design, Synthesis and Evaluation. *New J. Chem.* 2011, 35, 542-545.
11. Yaogang Hu, Xiaolong Li, Said M. Sebti, Jiandong Chen, and **Jianfeng Cai***. Design and Synthesis of AApeptides: A New Class of Peptide Mimics. *Bioorg. Med. Chem. Lett.*, 2011, 21, 1469-1471.

Work from Graduate and Postdoc.

10. Rongsheng E. Wang, Raj K. Pandita, **Jianfeng Cai**, Clayton R. Hunt, John-Stephen Taylor*. Inhibition of Heat Shock Transcription Factor Binding by a Linear Polyamide Binding in an Unusual 1:1 Mode. *ChemBioChem*, 2012, 13(1), 97-104.
9. Sourav Saha, **Jianfeng Cai**, Daniel Eiler and Andrew D. Hamilton*. Programing the formation of DNA and PNA quadruplexes by pi-pi stacking interactions. *Chem. Commun.*, 2010, 46, 1685-1687.
8. Yao Cheng, Lun K. Tsou, **Jianfeng Cai**, Toshihiro Aya, Ginger E. Dutschman, Elizabeth A. Gullen, Susan P. Grill, Annie Pei-Chun Chen, Brett D. Lindenbach, Andrew D. Hamilton, Yung-chi Cheng*. A novel class of meso-tetrakis-porphyrin derivatives exhibit potent activities against hepatitis C virus genotype 1b replicons *in vitro*. *Antimicrob. Agents Chemother.* 2010, 54(1), 197-206.

7. **Jianfeng Cai**, Dariusz Niedzwiedzki, Harry A. Frank*, and Andrew D. Hamilton*. Ultrafast energy transfer within pyropheophorbide-a tethered to self-assembling DNA Quadruplex. *Chem. Commun.* 2010, 46, 544 - 546.
6. **Jianfeng Cai**, Brooke Rosenzweig, and Andrew D. Hamilton*. Inhibition of Chymotrypsin by a selfassembled DNA quadruplex functionalized with cyclic peptide binding fragments. *Chem. Eur. J.*, 2009, 15(2), 328-332.
5. **Jianfeng Cai**, Erik M. Shapiro*, and Andrew D. Hamilton*. Self-assembled DNA quadruplex conjugated to MRI contrast agent. *Bioconjugate Chem.*, 2009, 20(2), 205-208.
4. **Jianfeng Cai**, Xiaoxu Li, and John Stephen Taylor*. Improved nucleic acid triggered probe activation through the use of a 5-thiomethyluracil peptide nucleic acid building block. *Org. Lett.*, 2005, 7(5), 751754.
3. **Jianfeng Cai**, Xiaoxu Li, Xuan Yue, and John Stephen Taylor*. Nucleic acid-triggered fluorescent probe activation by the Staudinger reaction. *J. Am. Chem. Soc.*, 2004, 126(50), 16324-16325.
2. Yun Lu*, **Jianfeng Cai** and Gi Xue. Molecular design of a soft interphase and its role in the reinforcement and toughening of aluminum powder-filled polyurethane. *J. Adhes. Sci. Technol.*, 2001, 15, 71-82.
1. **Jianfeng Cai**, Yun Lu*, Gi Xue and Wei Zhang. The reinforcement of Al filled Polyurethane system. *Mod. Plastics Proc. Appl.*, 1999, 11 (6), 10.

PATENTS (ISSUED and APPLICATIONS) (at USF)

27. Lei Wang, Zhigao Wang, **Jianfeng Cai**. Methods of Inhibiting Necroptosis. **2026**, CT/US26/17018.
26. **Jianfeng Cai**, Heng Liu, Donghui Zhu, Firoz Akhter. Peptidomimetics inhibiting Abeta in vitro and in vivo for the treatment of AD. **2025**, 63/842,324.
25. **Jianfeng Cai**, Heng Liu. Helical sulfonyl-AApeptides modulating Abeta oligomerization and cytotoxicity by recognizing Abeta helix. **2023**, 63/609,860.
24. Ioannis Spanopoulos, Ali Azmy, **Jianfeng Cai**, Xue Zhao, Anamika Mishra, Mina Sharabiani Bagherifard. Porous hybrid metal halide semiconductors. **2023**, PCT/US23/78399.
23. **Jianfeng Cai**, Sami Abdulkadir. Inhibition of Hypoxia-Inducible Transcription Factor (HIF-1alpha) Signaling with Sulfonyl-gamma-AApeptide Helices. **2023**, 24T026PR-CS.
22. Yu Chen, **Jianfeng Cai**, Prahathees Eswara , Michael Sacco, Lei Wang, Lauren Hammond, Hiran Malinda Lamabadu. Discovery of small molecular inhibitors against Staphylococcus aureus GpsB from protein-protein interactions. **2023**, 24T017PR-CS.
21. Niketa A. Patel, **Jianfeng Cai**. Small molecule targeting lncRNA in neurodegenerative diseases and tauopathies, **2023**, OI2023-01160.
20. **Jianfeng Cai**, Shibo Jiang, Lu Lu, Songyi Xue, Xinling Wang, Lei Wang, Wei Xu, Shuai Xia. Novel cyclic gamma-AApeptide-based long-acting pan-coronavirus fusion inhibitor with potential oral bioavailability by targeting two sites in spike protein. **2022**, 63/371,671.
19. **Jianfeng Cai**, Bo Huang, Jun Yin, Li Zhou and Ruochuan Liu. Activation of e6ap/ube3a-mediated protein ubiquitination and degradation pathways by a cyclic γ -AA peptide. **2022**, 63/266,907.
18. **Jianfeng Cai**, Sami Abdulkadir. Modulating angiogenesis by proteomimetics of vascular endothelial growth factor. **2021**, 63/265,835, **US 11,866,474 B2**.
17. **Jianfeng Cai**, Mengmeng Zheng. Discovery of cyclic peptidomimetic ligands targeting the extracellular domain of EGFR. **2021**, 63/202,564
16. **Jianfeng Cai**, Mengmeng Zheng. Peptidomimetic-based Antibody Surrogate for HER2. **2021**, USF Ref. No. 21A053PR-CS.
15. Chuanhai Cao, **Jianfeng Cai**, Compositions and methods relating to sulfono-gamma-aa peptides, **2021**, 63136903.

14. **Jianfeng Cai**, Chuanhai Cao. Novel compounds for the treatment of neurodegenerative diseases, **2019**, 16726575.
13. **Jianfeng Cai**, Peng Sang, Yan Shi, Haitao Ji, Min Zhang. β -catenin/B-cell lymphoma 9 protein-protein interaction inhibiting peptidomimetics, **2019**, 62/837,911.
12. Niketa A. Patel, **Jianfeng Cai**. Methods and compositions for diagnosis and management of neurodegenerative disease, **2018**, 62/515,727.
11. **Jianfeng Cai**, Peng Teng, Alekhya Nimmagadda. Novel bis-cyclic guanidines as antibacterial agents, **2017**, 62/536,295.
10. **Jianfeng Cai**, Yan Shi. One-Bead-Two-Compound Macrocyclic Library and Methods of Preparation and Use, **2017**, 62/483,038.
9. Vrushank Dave, **Jianfeng Cai**. PTEN Binding Compounds, Formulations, and Uses Thereof, **2017**, 62/460,324.
8. **Jianfeng Cai**, Ma Su, Alekhya Nimmagadda, Peng Teng. Cationic hydantoin compounds and the use of, **2016**, 62/426,698
7. **Jianfeng Cai**, Youhong Niu, Weibo Cai, and Hao Hong. RGD mimetic γ -AApeptides and methods of use. **2016**, US 9,234,007 B2, **issued**
6. **Jianfeng Cai**, Youhong Niu, Haifan Wu, Shruti Padhee. Identification of γ -AApeptides with potent and broad-spectrum antimicrobial activity. **2016**, US 9,499,587 B2.
5. Niketa A. Patel, **Jianfeng Cai**. Gas5 binding compounds, formulations, and uses thereof, 62/398,624, **2016**.
4. Said M. Sebt and **Jianfeng Cai**. Stapled peptides designed to inhibit the mutant KRas/ Raf interaction, **2016**, WO 172,187 A1.
3. **Jianfeng Cai**, Chuanhai Cao, Haifan Wu, Yaqiong Li, and Ge Bai. Methods of Synthesizing γ -AApeptides, γ -AApeptide Building Blocks, γ -AApeptide Libraries, and γ -AApeptide Inhibitors of Abeta40 Aggregates, **2016**, 0209422 A1.
2. Said M. Sebt, and **Jianfeng Cai**. Identification of Novel Inhibitors that Disrupt STAT3/DNA Interaction from γ -peptide OBOC Combinatorial Library, **2014**, Application No. 61/984179.
1. Nathan J. Rice, Lennox Hoyte, and **Jianfeng Cai**. Materials and methods for reliable measurement of blood volume. **2011**, PCT Int. Appl. WO 2011/30304.

BOOK CHAPTERS

7. Jarais Fontaine, **Jianfeng Cai**.* Application of Sulfonyl- γ -AApeptides for PPI Drug Discovery. Targeting Protein-Protein Interactions for Drug Discovery, 2026.
6. Jarais Fontaine, **Jianfeng Cai**.* Sulfonyl- γ -AApeptide tools for modulating biology. Methods in Enzymology, 2024.
5. Olapeju Oyesiku and **Jianfeng Cai**.* Peptidomimetic agents targeting bacteria. Comprehensive Supramolecular Chemistry II. Elsevier, 2017.
4. Peng Teng, Haifan Wu and **Jianfeng Cai**.* Peptidomimetics as antimicrobial agents. Novel Antimicrobial Agents and Strategies. Wiley, 2014.
3. Haifan Wu and **Jianfeng Cai**.* Engineering AApeptides for Translational Medicine. *Engineering in Translational Medicine*, 2013, ISBN: 978-1-62703-651-1.
2. Youhong Niu, Yaogang Hu, Haifan Wu, and **Jianfeng Cai**.* Synthesis of AApeptides. *Peptide Modifications to Increase Metabolic Stability and Activity*, 2013, ISBN: 978-1-62703-651-1.
1. Youhong Niu, Yaogang Hu, Rongsheng E. Wang, Xiaolong Li, Haifan Wu, Jiandong Chen* and **Jianfeng Cai**.* AApeptides as a New Class of Peptidomimetics to Regulate Protein-Protein Interactions. *Protein Interactions*, 2012, ISBN: 978-953-51-0244-1.

ORAL TALKS AND SEMINARS

1. Florida Organic Day, Florida Southern College, 03/12/2012
2. Florida ACS meeting, Tampa, FL, 05/09/2012
3. Kimberly-Clark, Appleton, WI, 06/02/2012
4. Department of Chemistry, University of Oxford, Oxford, England, 06/07/2012
5. Interventional Cancer Institute of Integrative Medicine, Putuo Hospital, Shanghai, China, 12/12/2012
6. Department of Chemistry, University of Florida, Gainesville, FL, 11/15/2013
7. Department of Chemistry and Biochemistry, University of California-Santa Barbara, Santa Barbara, CA, 2/27/2014
8. Department of Chemistry, University of California-Irvine, Irvine, CA, 2/28/2014
9. Department of Chemistry and Biochemistry, Georgia Institute of Technology, GA, 3/10/2014
10. Department of Chemistry, Georgia State University, Atlanta, GA, 3/11/2014
11. Department of Chemistry, University of South Florida, GA, 3/13/2014
12. 247th ACS national meeting, Organic section, Dallas, TX, 3/17/2014
13. Department of Chemistry, Florida State University, Tallahassee, FL, 3/27/2014
14. Department of Chemistry, University of Wisconsin-Madison, Madison, WI, 4/3/2014
15. Kimberly-Clark, Appleton, WI, 4/4/2014
16. Department of Chemistry, Scripps Florida, Jupiter, FL, 4/17/2014
17. Innovative Drug Research Center, Chongqing University, Chongqing, China, 5/6/2014
18. Department of Chemistry, Nanjing University, Nanjing, China, 5/7/2014
19. College of Pharmacy, Shanghai Jiaotong University, Shanghai, China, 5/8/2014
20. Department of Medical Oncology, Shuguang Hospital, Shanghai University of Traditional Chinese Medicine, Shanghai, China, 5/9/2014
21. Bioorganic Gordon Conference, Andover, NH, 6/11/2014
22. Department of Chemistry, Washington University in St. Louis, MO, 4/23/2015
23. Department of Chemistry, University of Missouri-St. Louis, 4/24/2015
24. Department of Chemistry, Southeast University, China, 6/25/2015
25. College of Pharmacy, Zhejiang University, China, 6/26/2015
26. Department of Chemistry, Central South University, China, 7/1/2015
27. Lawrence Berkeley National Laboratory, San Francisco, 8/6/2015
28. College of Medicine, University of South Florida, 9/16/2015
29. Department of Chemistry, UC-Riverside, 2/25/2016
30. Department of Chemistry, Dartmouth College, 4/14/2016
31. FAME 2016-Florida Annual meeting and Exposition, FL, 5/6/2016
32. Department of Chemistry, University of South Carolina, 3/30/2017
33. Department of Chemistry, University of South Dakota, 4/11/2017
34. FAME 2016-Florida Annual meeting and Exposition, FL, 5/6/2017
35. Department of Chemistry, Zhengzhou University, China, 5/9/2017
36. Department of Chemistry, Zhengzhou University of Light Industry, China, 5/9/2017
37. Department of Chemistry, Nanjing University, China, 5/10/2017
38. Department of Chemistry, China Pharmaceutical University, China, 5/11/2017
39. Department of Chemistry, Southeastern University, China, 5/12/2017
40. Department of Chemistry, Fudan University, China, 5/15/2017
41. Department of Chemistry, East China University of Science and Technology University, China, 5/16/2017
42. Department of Chemistry, Soochow University, China, 5/17/2017

43. Department of Chemistry, Central South University, China, 5/19/2017
44. Department of Chemistry, Hunan University, China, 5/22/2017
45. Department of Chemistry, Hunan Normal University, China, 5/22/2017
46. Department of Chemistry, Wuhan University, China, 5/23/2017
47. College of Pharmacy, Wuhan University, China, 5/24/2017
48. Department of Chemistry, Central China Normal University, China, 5/26/2017
49. Department of Chemistry, Shanxi Normal University, China, 5/17/2018
50. Department of Chemistry, Xi'an Jiaotong University, China, 5/18/2018
51. Department of Chemistry, Northwest University, China, 5/19/2018
52. Department of Chemistry, University at Buffalo, 9/12/2019
53. Department of Chemistry, Case Western Reserve University, 4/10/2019
54. Department of Chemistry, University at Albany, 9/10/2019
55. College of Pharmacy, University of Arizona, 1/8/2020
56. Department of Chemistry, Tulane University, 3/15/2021
57. Pacificchem 2021, Advancing Frontiers in Peptide and Protein Science with Nano- to-Macro Molecular Solutions, New Technologies in Polyamide Synthesis and Applications (056), 12/18/2021
58. Pacificchem 2021, Design of Functional Proteins, Peptides, and Peptidomimetics (061), 12/20/2021
59. Department of Chemistry, University of New Mexico, 04/29/2022
60. 28th American Peptide Symposium, Scottsdale, AZ, 06/26/2023
61. Foldamer 2023 symposium, Munich, 09/04/2023
62. The Huber and Helen Croft Lectureship, University of Missouri-Columbia, 09/22/2023
63. Virginia Commonwealth University, 12/14/2023
64. College of Pharmacy, University of Maryland, 03/06/2024
65. University of Texas-Arlington, 04/04/2024
66. College of Pharmacy, Purdue University, 04/25/2024
67. Pharmacology Program, Mayo Clinic, 05/11/2024
68. 2025 Fall ACS meeting, Washington D.C., 08/18/2025

ACTIVE GRANTS

As the PI:

1. PI, NIH 2R01AG056569-06, \$2,902,465, 03/01/2023-12/31/2027, Recognition of Abeta monomeric helix.
2. PI, NIH 1R01GM150196, \$1,661,240, 04/01/2023-03/31/2027, Targeting Wnt signaling pathway.
3. PI, NIH 1RF1AG094856-01, \$3,297,926, 09/01/2025 – 08/31/2029, Inhibition of Abeta42 aggregation by helix stabilizing peptidomimetic biomaterials.

As the Co-PI or Co-I:

1. Co-I, NIH 2R42AG078718-02A1, (\$360,000 to Cai), 09/20/2025-05/31/2028, Development of "Imagnostix": A Liquid Biopsy System for Alzheimer's Disease.